SIGNA Voyager

Software Upgrade Instruction (PX26.0/PX26.2 to VX28.0)

5836770-1EN
Revision 2.0

Language Policy

Direction 2128126 - Language Policy For Service Documentation

ПРЕДУПРЕЖ ДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. • Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. • Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. • Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 • 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤 害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 • 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 • 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 • 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他的危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 • 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 • 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 • 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZOR-ENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. • Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. • Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. • Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. • V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. • Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. • V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. • Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. • Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. • Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAAR-SCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. • Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. • Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. • Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. • Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Τοπαρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. • Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. • Μηνεπιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. • Εάν δεν λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. • Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. • Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu. • Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. • Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. • Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 • サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 • このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者 さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. • 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. • 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. • Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării şi înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanemarivanje ovog upozorenja može dovesti do povređivanja servisera, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här service-handboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. • Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. • Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. • Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.
ЗАСТЕРЕЖЕННЯ (UK)	Даний посібник з експлуатації доступний тільки англійською мовою. • Якщо постачальник послуг клієнта спілкується іноземною мовою, тоді клієнт зобов'язаний забезпечити переклад. • Заборонено проводити огляд обладнання без попереднього звертання до даного посібника з експлуатації і розуміння інформації, поданої у ньому. • Недотримання цього застереження може завдати шкоди здоров'ю постачальника послуг, оператора або пацієнта через ураження електричним струмом, механічну травму або інше ушкодження.

Revision History

Rev	Rev	Rev
1.0	Jan 3, 2020	Initial Release.
2.0	Jan 10, 2020	Formal Release for M3.

Contents

Page intentionally left blank

1 Getting Started

1.1 SIGNA Voyager Software Upgrade Instruction (PX26.0/PX26.2 to VX28.0)

1.1.1 Overview

This instruction applies to all ***SIGNA Voyager*** sites installed with PX26.0/PX26.2 software in the install base.

1.1.2 Purpose

The upgrade from PX26.0 / PX26.2 to VX28.0 on SIGNA Voyager product with enhanced software and application feature (purchasable).

1.1.3 Furnished Material in Upgrade Kit

Table 1 Kit Part#: M70073AE

GE Part Number	Description	Qty
5835373	Voyager IPMAir Software Collector	1
6339325-199	SIGNA Voyager Operator Manual Booklet	1
6339225-399	SIGNA Voyager Operator Manual	1
5836770-1EN	SIGNA Voyager Software Upgrade Instruction (PX26.0/ PX26.2 to VX28.0)	1
5732368	SIGNA Voyager Service Manual Collector	1

1.1.4 Disposition of Material

Retain the Upgrade Kit at site until the next version of software is provided.

Page intentionally left blank

2 USB SW Installation and RSVPReference topics

2.1 Reloading or upgrading USB host system software

Personnel requirements			
Required persons	**Preliminary require-ments**	**Procedure**	**Finalization**
1	-	90 minutes	5 minutes

Tools and test equipment			
Item	**Quantity**	**Part number**	**Manufacturer**
USB Software	1	-	-
Service Pack DVD (if appli-cable)	1	-	-
Saveinfo CD/DVD	1		
GE Equipment Mainte-nance Sticker	1	5661793	-

Required conditions
All media except for Save Info DVD must be removed during the loading of the application software, disk parti-tioning, and application software load. This includes other DVDs, MODs, and USB devices (for example, service keys). Failure to remove other media can cause software load failure.
System monitor turned on.
All USB devices (including service keys) are removed.
Any non-OS media is removed.

Safety
Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare *MR Service Safety Manual* (5452735). • Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

1. Insert the USB software device required for this procedure.

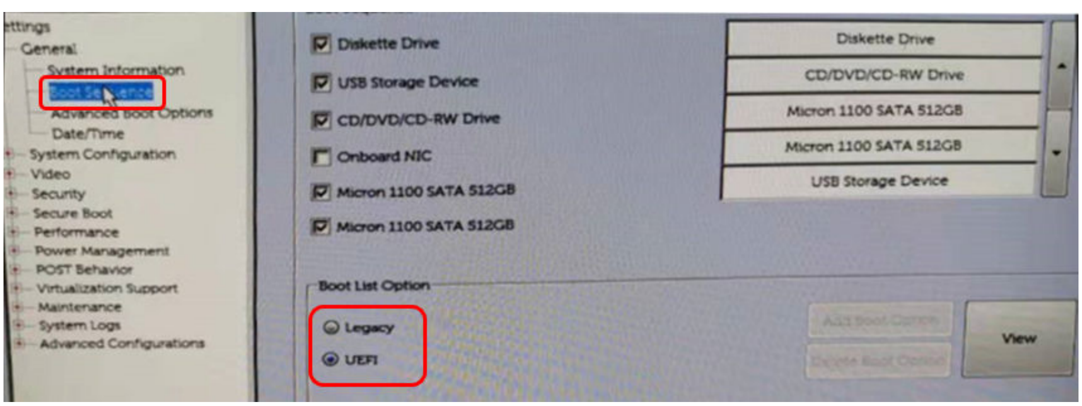
   NOTE

   A USB-3 port is recommended. The port has a blue-colored plug or is labeled on the box.

   Figure 1 USB-3 port symbol

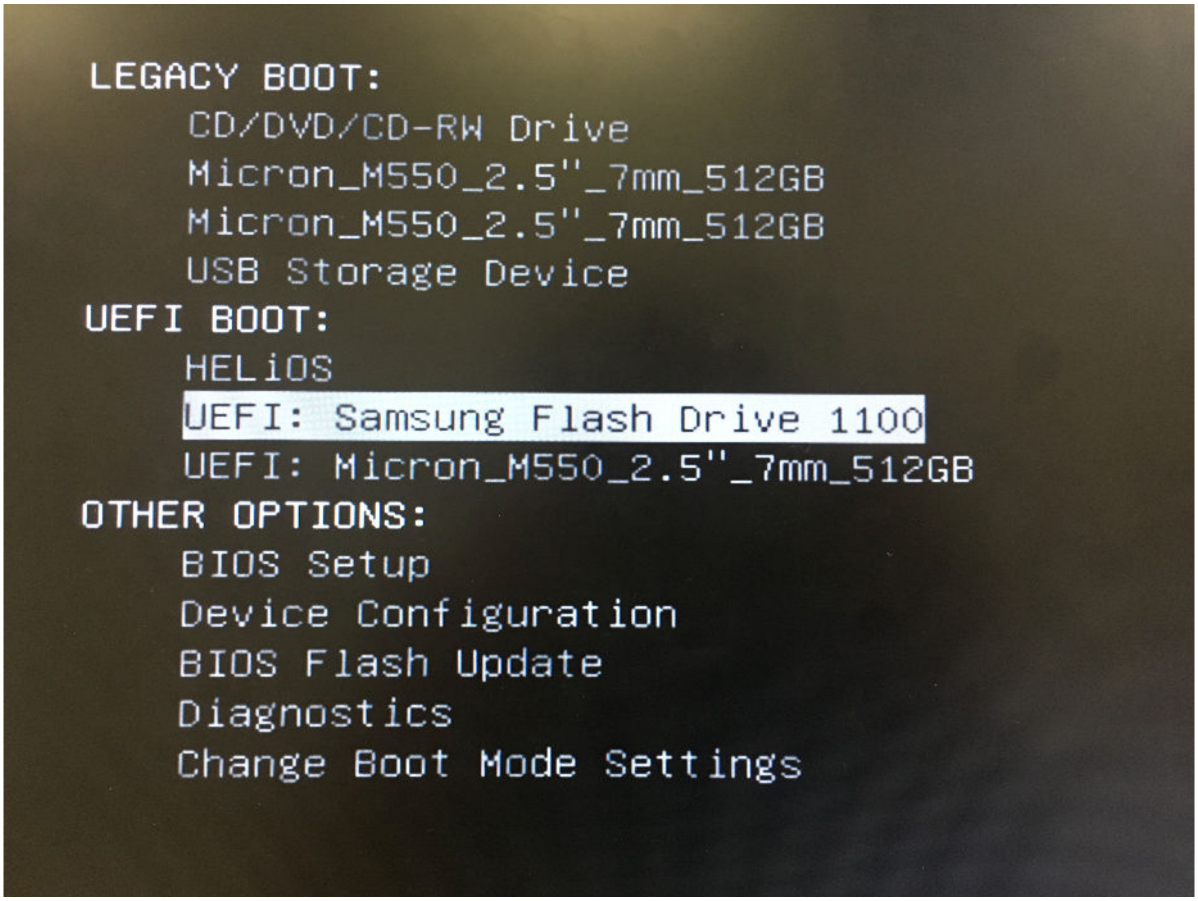
2. Setup the correct Firmware on the host PC:
   a. Press the ***Power*** button on the host PC to boot the system.
      i. Press ***F12*** to enter the ***Boot*** menu.
      ii. Make sure the Boot mode is set to UEFI. If the Boot mode is set to Legacy, you must change it to UEFI mode. See Changing Boot mode.



3. Insert the SaveInfo CD/DVD to the DVD drive.

4. Use the up and down arrows on the keyboard to select the USB software media listed under ***UEFI Boot***. Brand and model may vary.

Figure 2 UEFI Boot manu



5. Select the software revision to install and start the software load from Boot loader. The format of the option displayed is: `[apps]<Apps Software revision>`.

6. OS installation:

NOTE

At times during the software load, a blank (frozen) screen may appear due to a known Red Hat bug (bug ID 1432436). If this occurs, you must restart the software load from the beginning. Press ***Ctrl*** + ***Alt*** + ***F2*** to access the Command prompt screen and restart the software load.

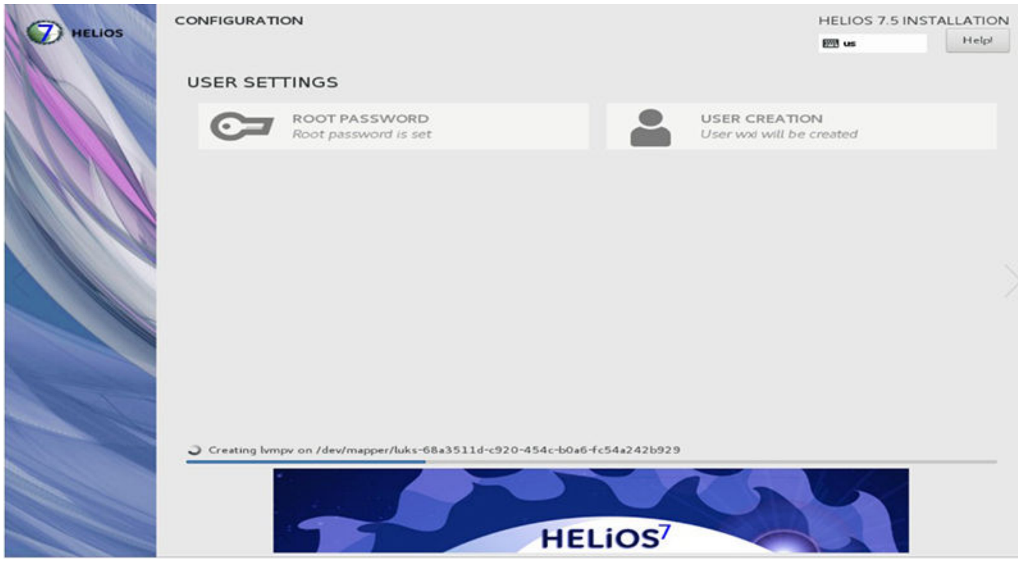
Figure 3 Blank screen due to Red Hat bug

- At the start of OS installation, automated OS confiuration takes place and then OS install starts within a few minutes.

Figure 4 OS installation - start

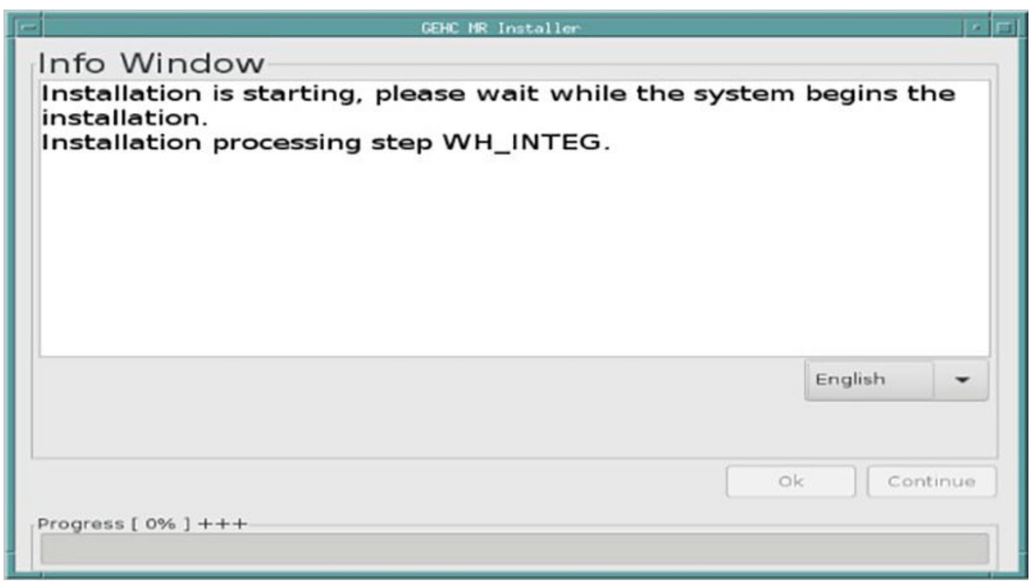
- OS installation continues to completion.

Figure 5 OS installation - completion



7. If the OS install fails, remove and re-insert the USB software device, and try the install again.
   - If the OS install is successful, the system automatically reboots.
   - The applications software load automatically starts. Please check the install progress according to the progress bar at the bottom of the window.

Figure 6 Applications software load - start



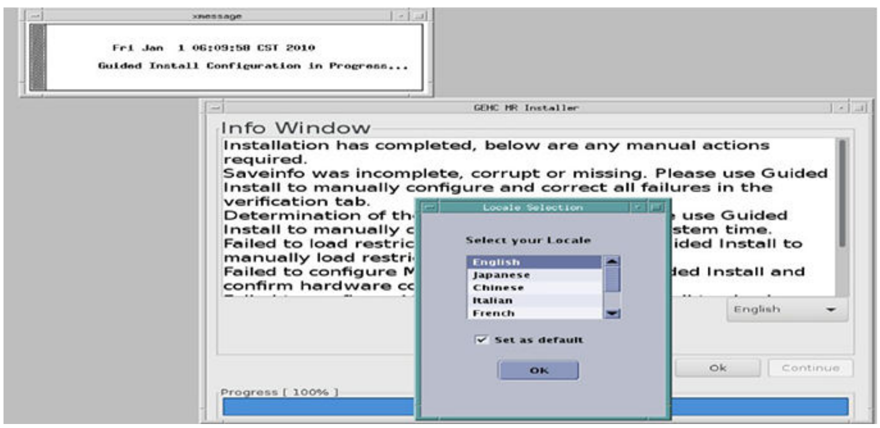
   - The language menu can be used to select a different language , if needed.

8. If the applications software install fails for any reason, remove and re-insert the USB software device, and try the install again.
    - ICN configuration starts automatically. A Java window appears showing the ICN software load progress.
        - To view the installation progress window, click ***Accept*** > ***Run***.
        - To bypass the installation progress window, click ***Cancel*** > ***Cancel***.
    - The Installer Info window appears along with the Locale Selection window. Choose "Set as default" and click "ok" to proceed the procedure.

    NOTE

    The Info Window lists all the software load actions that have either not been attempted or have failed.

    Figure 7 Installer Info and Locale Selection

9. Guided Install language selection and System Configuration should now be completed. Click Configure on each tab of Guided Install where configuration is updated.
10. Check the hardware configuration (VX28.0).

    Product name 1.5T, 70cm, SIGNA Voyager

    Field Strength Set to 1.5T

    Package Name Set to SSSD

    Gradient type Set to 5550(SSSD)

    Resonance Module Set to VRMW

    Magnet Serial Number Enter actual magnet serial number; first two letters are RD or PM

    Magnet Ramp Direction Default

    Table Limit Default; cannot change

    Scan Range (micro-meter) Set to long

    Line Frequency Set to 50 or 60 Hz

RF Amp Type Set to 1.5T 16KW XRFD

Governing Body Select proper governing body for the customer site; usually iec (or other selection if required by local regulation)

ISO Vector Z Set to 1.5T 70cm magnet enclosure

Magnet Enclosure Set to SIGNA Voyager

PAC type Select PAC2B/PAC5

RF Systems Cabinet (RFS) Select Yes

Spectro RF Amp type Select None if site does not have the spectroscopy option, or AMT if site does have spectroscopy option

Scanner channel configuration Set to 33/49/65

NOTE

FE should select the channel number according to option key customer ordered.
If customer has M7002AR or M7002AJ (49 Channels Upgrade), 49 channels should be selected;

If customer has M70012TM or M70072AQ (65 Channels Capability), 65 channels should be selected.

Otherwise, 33 channels (default value) should be selected.

Shim supply type Default

LPCA/port configuration type Select 3 P-Ports + PA-Port

Cooling cabinet Type Default

UPM Type Default; QUPM 1MHz sampling rate

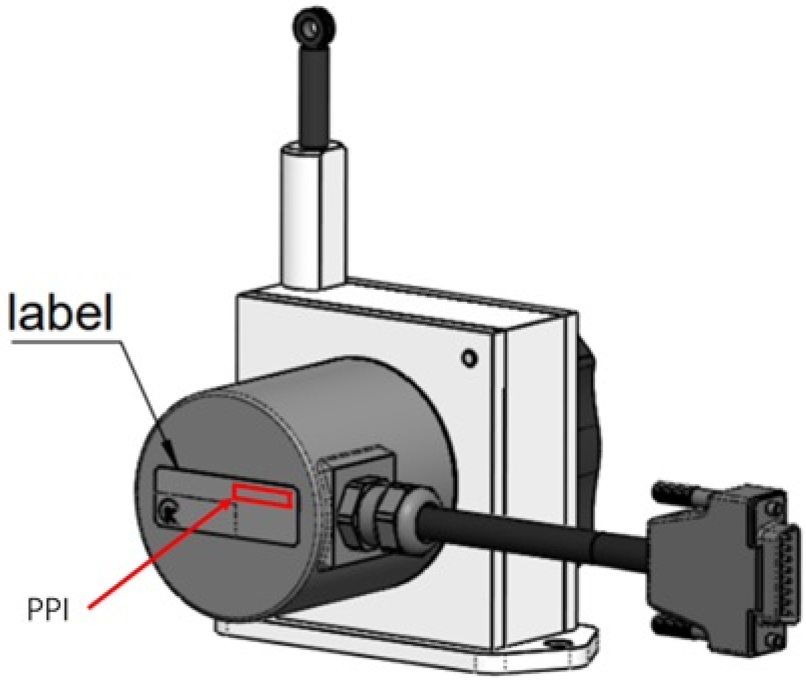
Table configuration Select Fixed table

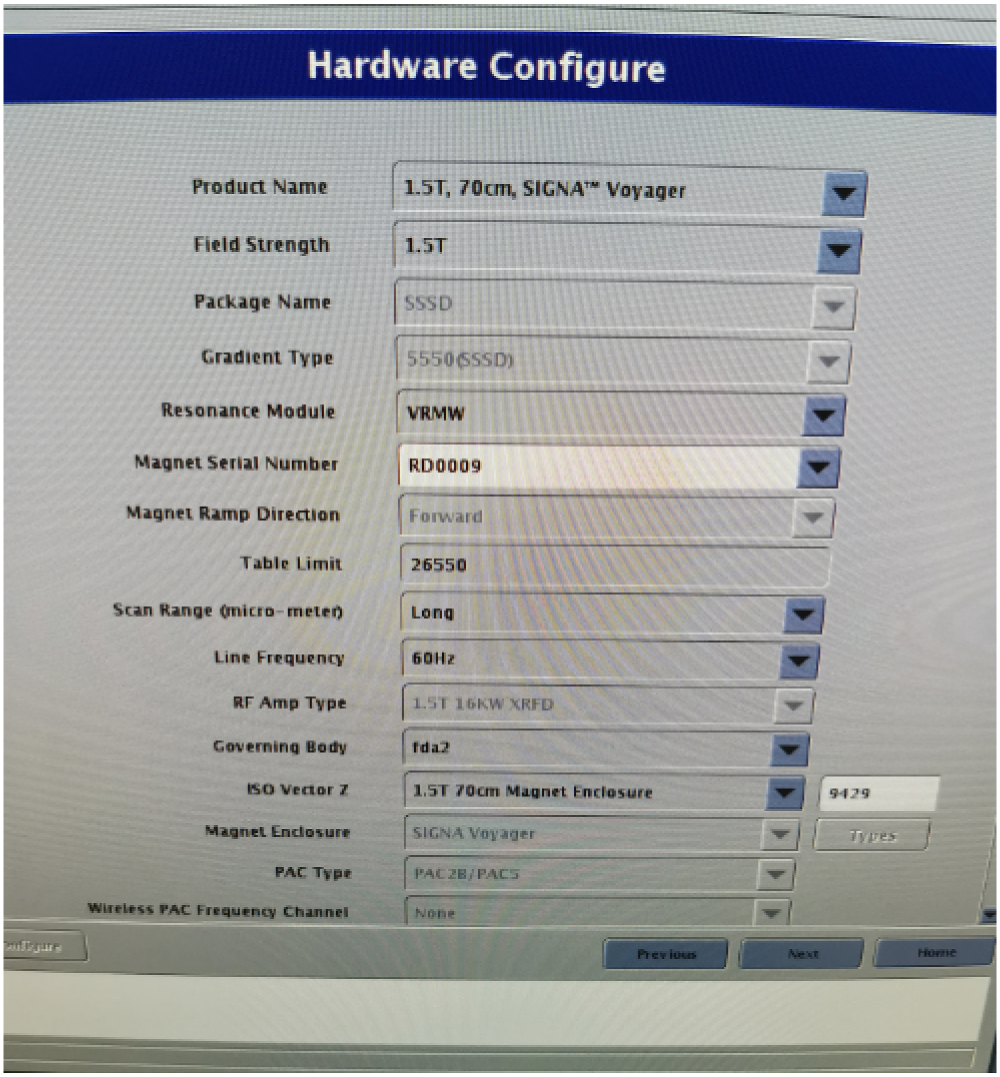
SSC Type Select ICE

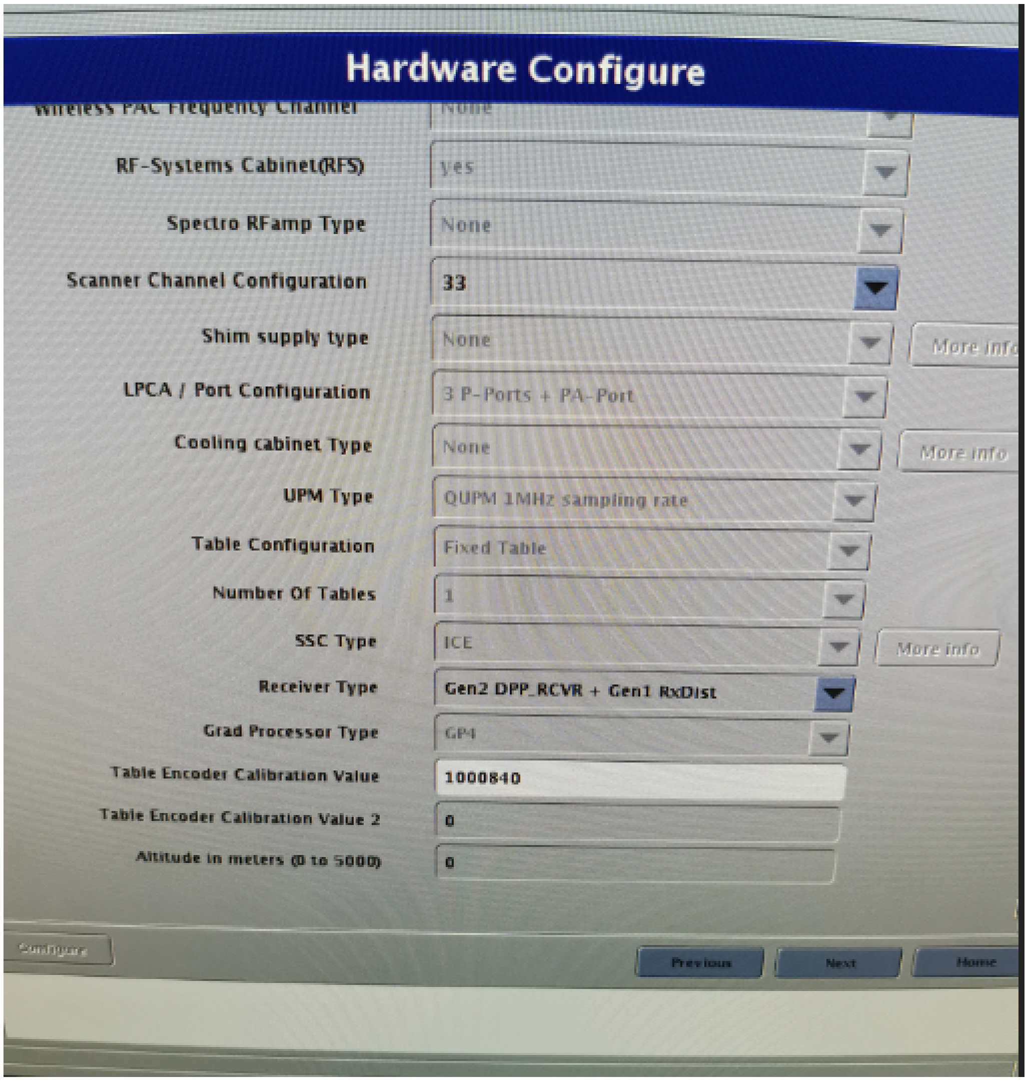
Receiver Type Select Gen2 DPP_RCVR + Gen 1 RxDist

Grad Processor Type Default; GP4

Table Encoder Calibration Value Input the value to displayed the encoder label. See below figure


label
PPI


Hardware Configure
Product Name
1.5T, 70cm, SIGNA™ Voyager
Field Strength
1.5T
Package Name
SSSD
Gradient Type
5550(SSSD)
Resonance Module
VRMW
Magnet Serial Number
RD0009
Magnet Ramp Direction
Forward
Table Limit
26550
Scan Range (micro-meter)
Long
Line Frequency
60Hz
RF Amp Type
1.5T 16KW XRFD
Governing Body
fda2
ISO Vector Z
1.5T 70cm Magnet Enclosure
9429
Magnet Enclosure
SIGNA Voyager
Types
PAC Type
PAC2B/PAC5
Wireless PAC Frequency Channel
None
Previous
Next
Home


Hardware Configure
RF-Systems Cabinet(RFS)
yes
Spectro RFamp Type
None
Scanner Channel Configuration
33
Shim supply type
None
LPCA / Port Configuration
3 P-Ports + PA-Port
Cooling cabinet Type
None
More info
UPM Type
QUPM 1MHz sampling rate
Table Configuration
Fixed Table
Number Of Tables
1
SSC Type
ICE
More info
Receiver Type
Gen2 DPP_RCVR + Gen1 RxDist
Grad Processor Type
GP4
Table Encoder Calibration Value
1000840
Table Encoder Calibration Value 2
0
Altitude in meters (0 to 5000)
0
Previous
Next
Home

11. Check the ***Verification*** tab of Guided Install to make sure all entries pass verification.

    NOTE

    Pending software load tasks from the Info Window must be completed manually. To complete the actions, see Loading USB Host System Software (Quick Start)(Refer to SM 5680005-2EN). The system will not boot to a scannable state if these actions are not completed. Pending tasks include:

    - Restore Info (if Restore Info failed)
    - Guided Install Configuration entries (based on Restore Info status)
    - ICN Configuration
    - IRM installation (if Restore Info failed)
    - Service Pack Load as necessary per DOC1667089
    - Service Software Load (this task is not automated as part of the USB software load)
    - Setup InSite RSVP Connectivity (if Restore Info failed)
    - SSA License installation (if Restore Info failed)
    - Option Keys installation (if Restore Info failed or new options are purchased)
    - Operator Manual installation
    - Reboot the system
    - Finalization steps as listed in Quick Start procedure (ICW actions)

    For detailed information on completing these remaining tasks individually, see *Installing host system software (Quick Start)*.

12. After all pending actions are complete, properly close Guided Install by selecting ***File*** on the top left corner of GUI and click on ***Quit***.
13. Close the GEHC MR Installer status window.

    A software check for upgrade scenarios runs, checking for software upgrade scenarios and sets up ICW as needed.

14. Return to *Installing host system software (Quick Start)* and complete the remaining tasks individually based on status of the software load.

2.2 Configuring the ICN - VRE

Volume Recon Engine (VRE) configuration reinstalls the software onto the Image Compute Node (ICN). This procedure is also needed after ICN replacement.

This step is only necessary if VRE needs to be replaced or if VRE configuration failed on previous step.

Personnel requirements			
Required persons	**Preliminary requirements**	**Procedure**	**Finalization**
1	10 minutes	30-60 minutes	5 minutes

Tools and test equipment			
Item	**Quantity**	**Part number**	**Manufacturer**
M-Class SSA Hard Key	1	-	-
MR VRE Linux OS DVD	1	-	-
MR Software USB Media	1	-	-

Safety
Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare *MR Service Safety Manual* (5452735). • Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

Volume Recon Engine (VRE) and Image Compute Node (ICN) are terms that are used interchangeably. VRE configuration installs the software onto the ICN. This procedure is needed after ICN replacement. This procedure may also be needed when troubleshooting issues with the VRE system. Refer to *Troubleshooting the VRE* for more information.

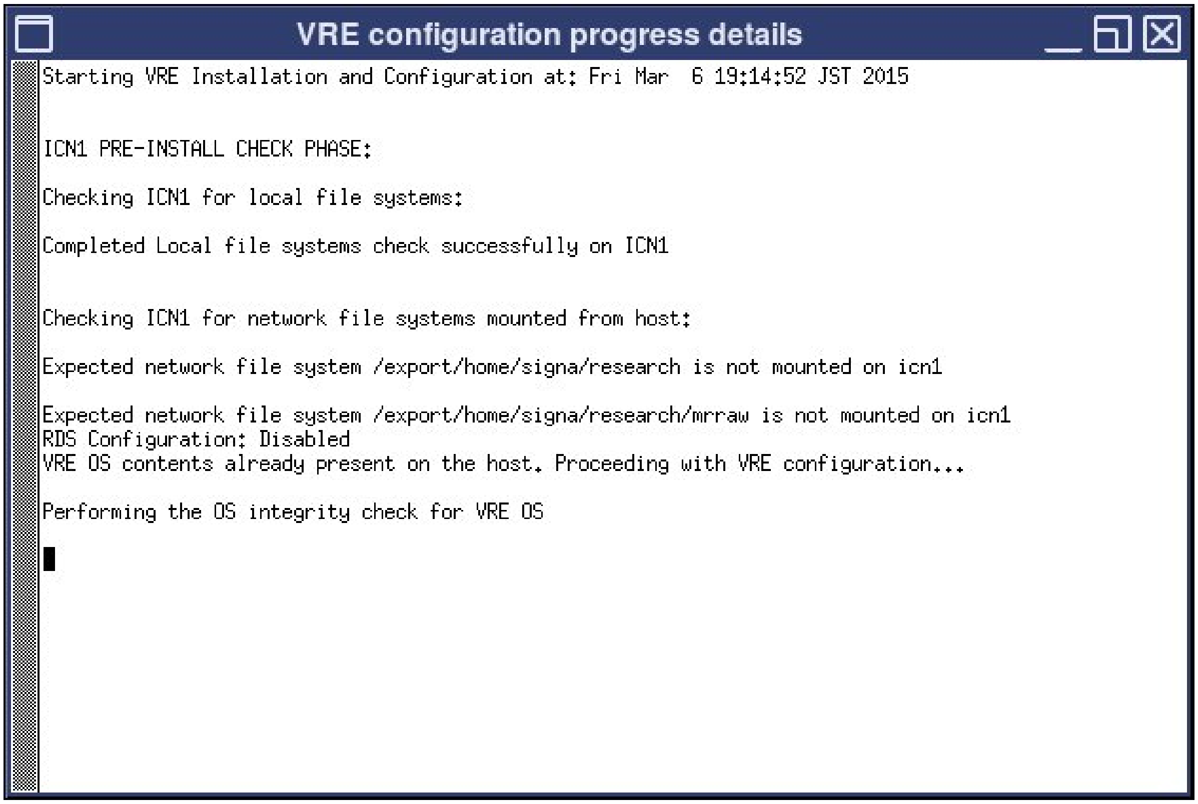
1. **NOTE**

   If you are replacing the ICN, power on the system and the host PC, and log in as root (password: operator).

   On the ***Tools*** screen, click the ***Service Desktop Manager*** tab.
2. Click ***Guided Install*** and then click ***Start***.
3. When prompted, enter the root login.
   It is possible that the customer changed the default password. If you cannot log in, contact the customer for the correct password.
4. Click ***Configure ICN***.
5. If prompted, insert the OS DVD, or USB as applicable , that came with the system software into the host PC and click ***OK*** to continue the ICN configuration (this step is only necessary if repository files are corrupted on the host computer).

6. If ICN configuration fails, review the dialog box and details in the separate window to determine next steps. See Troubleshooting the Image Compute Node (ICN).

Figure 8 ICN progress details

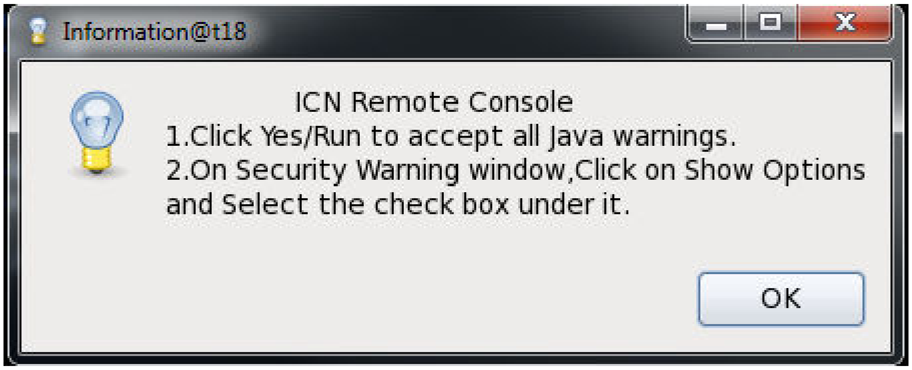


NOTE

For software version DV28/RX28/PX28 and later, when the Gen 6 ICN is being configured, a new ICN firmware maintenance step will follow. This is an automated step and is necessary to overcome LCC firmware issues that could lead to ICN configuration failures on the upgrade. For more details on the Firmware upgrade process refer to ICN Dell R630 (Gen6) firmware upgrade.

7. Once the ***ICN Remote Console*** window appears, click ***OK***, and then ***Continue***.
This will install the Java security options.

Figure 9 ICN remote console



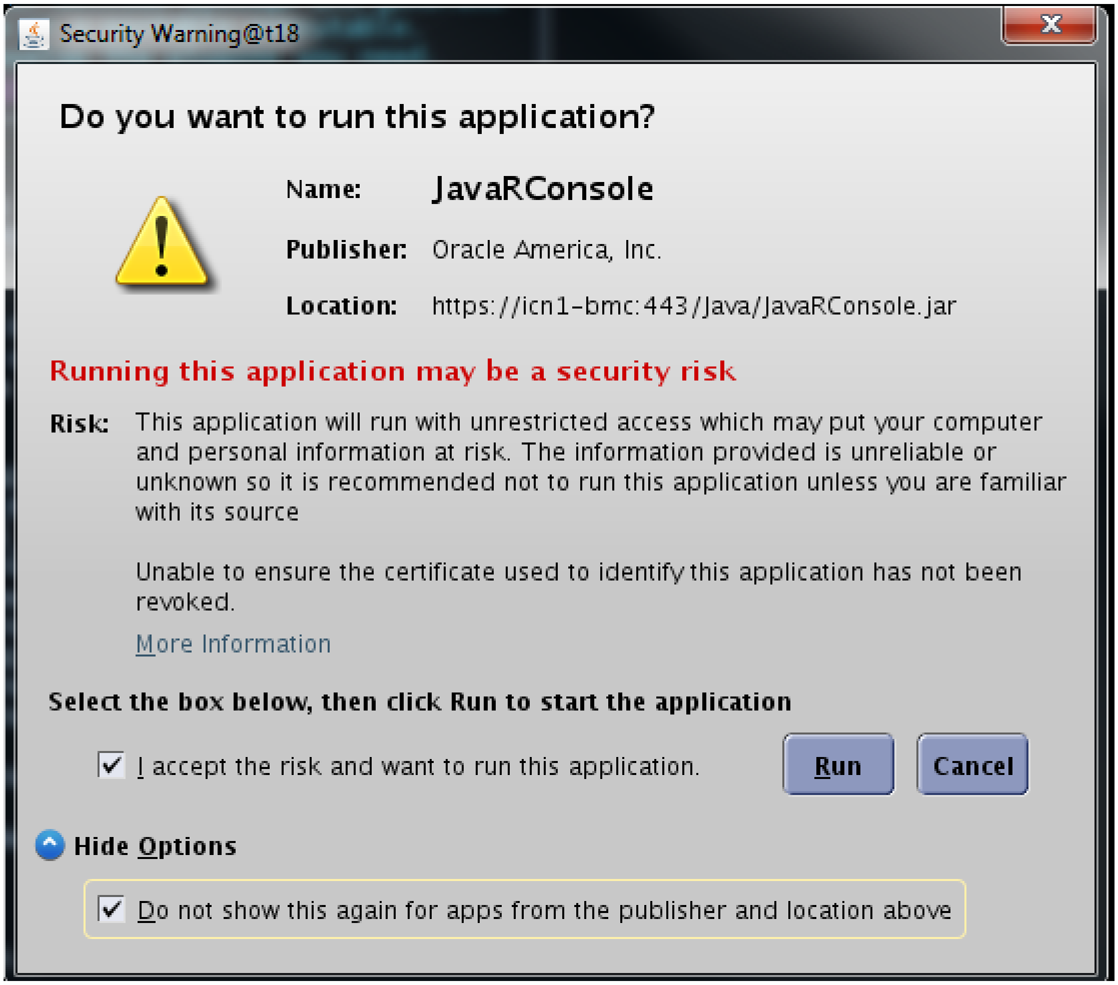
NOTE

This remote display will not launch if a valid service license is unavailable. A message shows that says missing service licenses will appear in the ICN configuration log. Ignore this message if you do not intend to look at the ICN remote display.

8. Once the Java ***Security Warning*** window appears select the following:
   a. Select the ***I accept the risk and want to run this application*** checkbox.
   b. Click on ***Show Options*** and select the ***Do not show this again for apps from the publisher and the location above.*** checkbox.

a. Click ***Run***.

Figure 10 Security Warning



The ***Remote Console*** window opens showing the progress of the installation.

9. At the end of VRE configuration, a dialog box displays with either a success or failure message. If VRE configuration fails, review the dialog box message and details in the separate window to determine next steps.

NOTE

On the ICN configuration screen, you can click ***View VRE Configuration file*** to view the output status of the ICN.

10. Click ***OK*** twice for the VRE configuration success and ICN configuration success messages.

2.3 Loading or Deleting Proprietary Software

Table 2 Personnel requirements

Required persons	Preliminary requirements	Procedure	Finalization
1	Not Applicable	30 minutes	Not Applicable

Table 3 Consumables

Item	Quantity	Effectivity	Part number	Manufacturer
MR Service Software CD/DVD or USB	1	-	-	-

This procedure explains how to install and uninstall proprietary and advanced service software. An installation GUI accessed from the Common Services Desktop directs the installation or removal of this software.

The ***MR Service Software CD/DVD or USB*** combines the ***Restricted Software*** and the ***Advanced Software*** into a single CD/DVD or USB and can be used to install either the Restricted Software or the Advanced Software, based on SSA licensing.

1. Loading Proprietary Software. Shut down all applications before loading the Restricted service or Advanced service media. This media should not be loaded with applications software running.
   a. Restart the system and log in as `root` with the password: `operator`

      NOTE

      It is possible that the customer changed the default password. If you cannot log in, contact the customer for the correct password.

      NOTE

      If continuing from Host Software Load, it is not necessary to reboot at this time.

   b. Open the Guided Install and select ***FE Mode*** and the link: ***Click here to start this tool***.

      NOTE

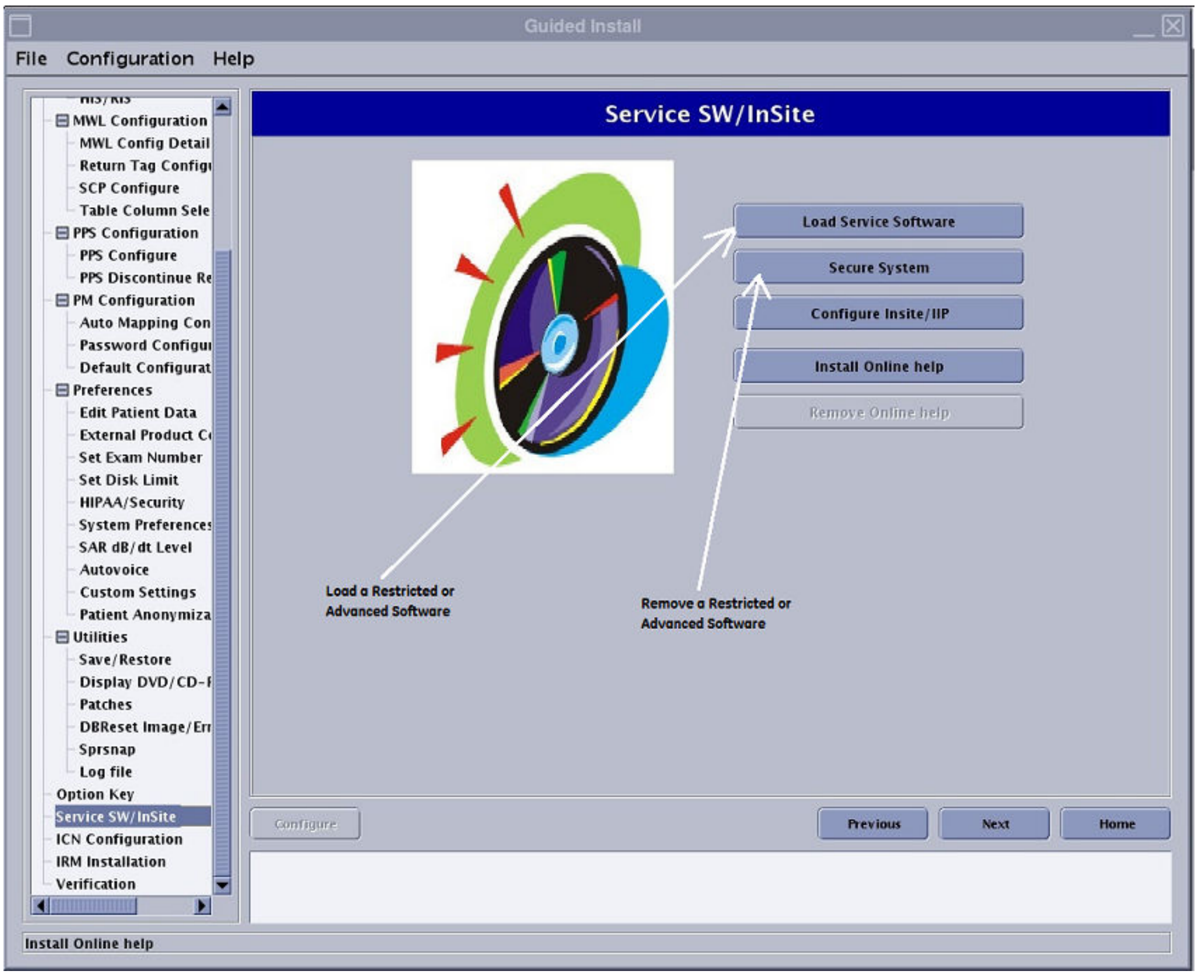
      Restricted service software is only available for GE use; Advanced service software is only available to sites with a valid Advanced Service Package Limited License. GE Healthcare Restricted service software permits configuration of InSite Interactive. Whether Restricted Software or Advanced Software installs is based on SSA licensing.

   c. Insert a valid SSA Class M hardkey at this time. This licensing mechanism is required to continue the Service Software Load. Continue to

d. Download and install the customer SSA soft licenses. This licensing mechanism is required to continue the Service Software Load.

2. Insert the MR Service Software CD/DVD or USB and from the left navigation menu, select ***Service SW/InSite*** and click on the ***Load Service Software*** button. The MR Service Software CD/DVD or USB was shipped together with the MR Application Software Collector.

Figure 11 Installing or Removing Proprietary Software



3. Deleting Proprietary Software

   To delete proprietary software, proceed as follows:

   - Open Guided Install and select ***FE Mode*** and the link: ***Click here to start this tool***.
   - From the left navigation menu of the Guided Install, select ***Service SW/InSite***.
   - Click ***Secure System*** to delete proprietary software.

4. Exit the Install GUI once completed

If proprietary and restricted software is loaded, configure the InSite Interactive Platform and complete the InSite Checkout. For details on the InSite Interactive platform, refer to Insite/IIP

2.4 Installing IRM software

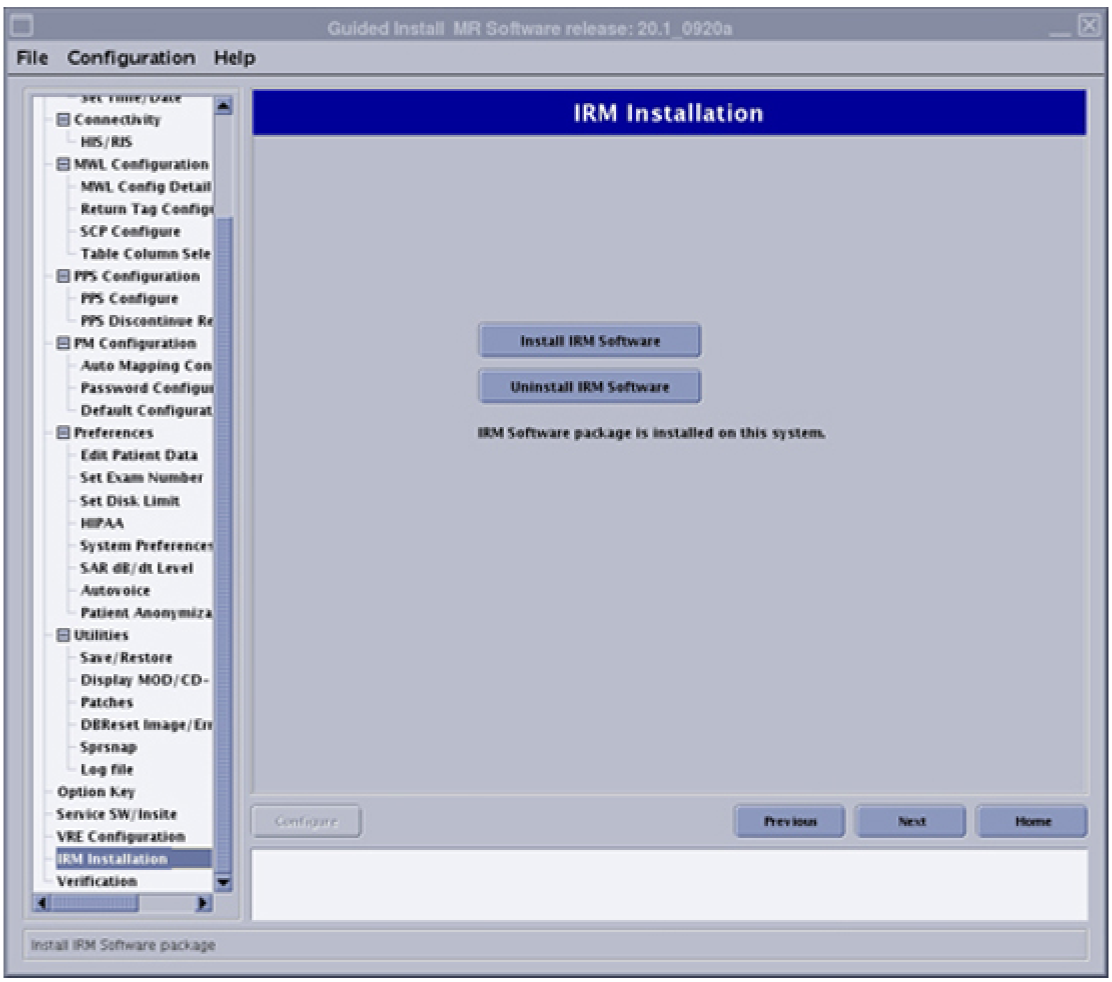
Follow this procedure if the site has an in-room monitor (IRM).

Follow this procedure if the site has an in-room monitor (IRM).

Personnel requirements			
Required Persons	**Preliminary require-ments**	**Procedure**	**Finalization**
1	-	5 minutes	-

1. Select ***Guided Install*** > ***IRM Installation***.
2. Click ***Install IRM Software***.

Figure 12 IRM installation screen



3. Click ***OK***.

2.5 Installing operator documentation (online help)

Load the operator manuals onto the host computer so customers can have access to the manuals any time they are using the system.

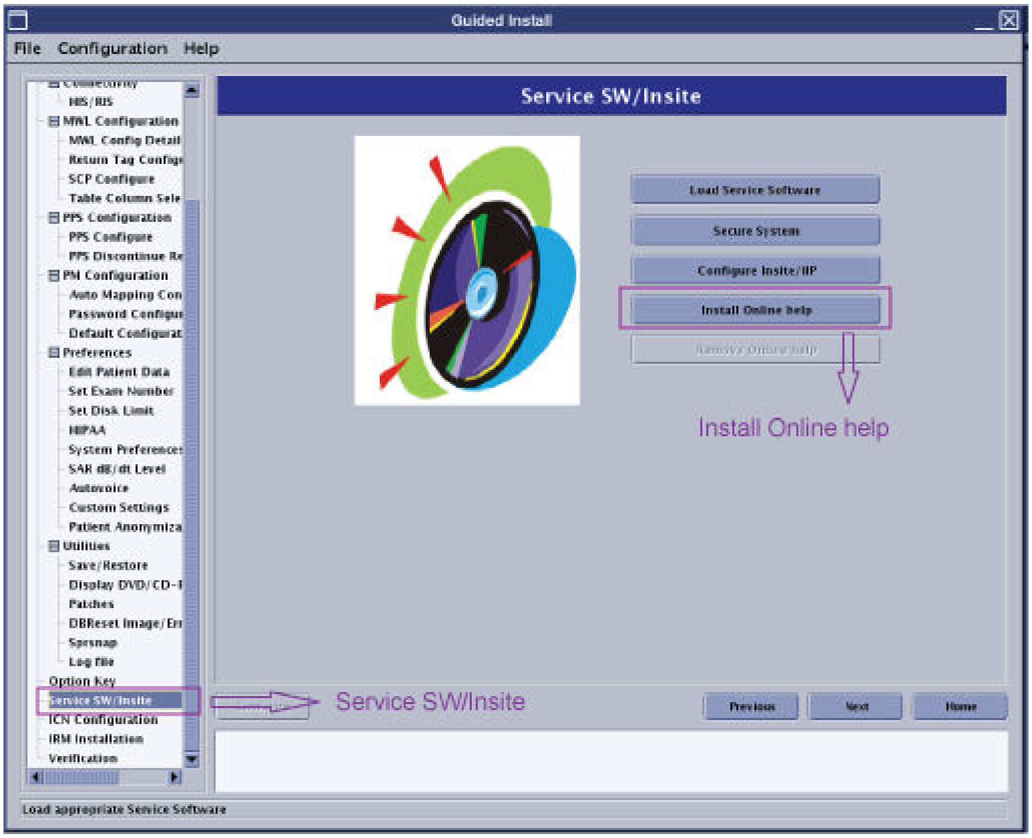
Personnel Requirements			
Required persons	**Preliminary require-ments**	**Procedure**	**Finalization**
1	-	10 minutes	-

Tools and test equipment			
Item	**Quantity**	**Part number**	**Manufacturer**
DVD Media for Operator Manual	1	-	-

1. Select the DVD that contains the language for the documentation that is to be installed on the system.
   The Operator Manual DVD is available in multiple languages. Each DVD label has the list of languages available.
2. Launch Guided Install. See Accessing Guided Install on page 39 for more information.

3. Select ***Service SW/Insite*** from the ***Guided Install*** content window.

Figure 13 Guided Install Service SW/Insite



4. Click ***Install Online Help***.

5. When prompted to insert the media containing online documentation, insert the DVD containing the selected language (DVD Media for Operator Manual) into the host computer's DVD drive and then click ***OK***.

NOTE

The DVD Media for Operator Manual is available in multiple languages. Verify that the selected DVD contains the language to be installed at the site.

An Online Help Installation dialog box appears.

6. Select the language that is to be used for the online help and click ***Install***.

Figure 14 Select language for online help

If the language you want is not on the menu, try another operator manual DVD.

After the installation is complete, a message appears indicating that the installation was successful.

7. Remove the DVD from the drive.
8. Reboot the system and login to applications.
9. To verify the online help opens, click the book icon on the lower right corner of the monitor and view the documents.

Figure 15 Online help icon

2.6 Perform ICW

1. Right-click on the screen background and select ***System Restart***. Restart the system and boot to applications as ***sdc***.

NOTE

The initial boot on a new system can take up to 15 minutes.

2. Once systems boots up completely, Launch Common Service Desktop (CSD) from the Svc Tools Desktop

NOTE

Since the CSD is being launched for the first time, a java pop up will appear. Please check the check box against “Do no show this again for apps from the publisher and location above” and click on ***Run***.

Figure 16 Java pop up

3. Launch ICW from the calibration tab and then complete the configuration. Refer to Configuring About MR Scanner in appendix.
4. Perform all the calibration which listed in upgrade mode.
   - If upgrade PX26.0 to VX28.0, Run About MR Scanner, LVShim-GradShim and SPT Full Test in calibration list. Refer to the procedure in appendix.
   - If upgrade PX26.2 to VX28.0, Run About MR Scanner and SPT Full Test in calibration list. Refer to the procedure in appendix.

2.7 InSite Remote Service Platform (RSvP)

Configuration information for the InSite Remote Service Platform (RSvP).

InSite RSvP replaces existing InSite IIP (InSite1) as the connectivity platform going forward. InSite RSvP is an enhanced connectivity platform based on InSite2 architecture (currently used by MM3 connectivity). The system will call the back office and identify/register itself. The following describes the essential steps required to configure the system functions.

2.7.1 InSite Remote Service Platform (RSvP) pre-installation requirements

Describes the configuration and information required prior to the installation of the Remote Service Platform (RSvP).

2.7.1.1 RSvP pre-installation requirements

The MR scanner requires connectivity directly to the Internet.

- The MR system allows for DNS configuration or proxy server-based connection to the Internet.
- A current Internet connection supporting InSite1 connection can be reused.
- RSvP uses SSL/TLS protocol, https over port 443.
- In rare cases when the customer does not accept the above connection protocol, after the issue is escalated, the local connectivity team can provide help to connect through SSL/TLS proxy IP over the site-to-site VPN.

The following information is required from the customer prior to starting an upgrade to RSvP:

- Final system ID for the site (final registration in FFA).
- DNS IP address or proxy IP address (required for modality configuration / Internet access).
- If a customer wants to only whitelist the specific URLs, the following are the required URLs for RSvP connectivity:
    - https://insite.gehealthcare.com
    - https://as1-insite.gehealthcare.com
    - https://as2-insite.gehealthcare.com
    - https://gehealthcare-ns.flexnetoperations.com
    - https://download.flexnetoperations.com

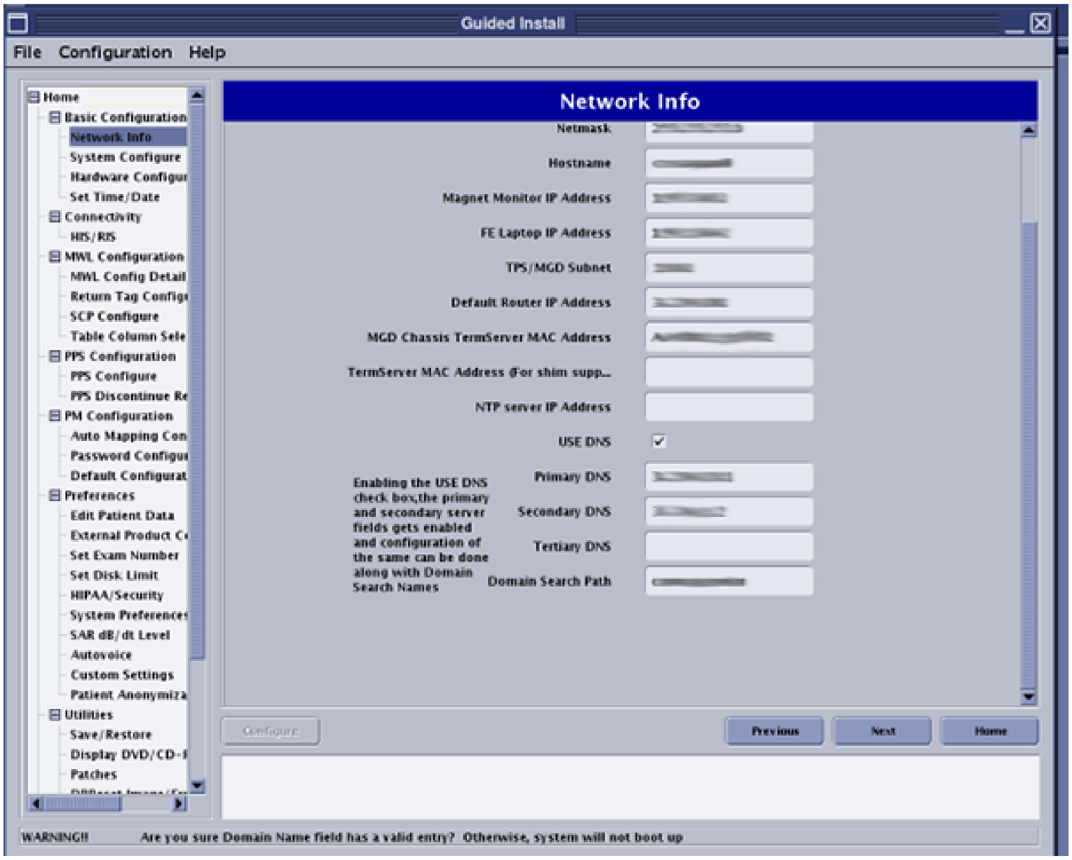
If the software being upgraded to RSvP is already connected through InSite1, do 2.7.2 Checking Internet connectivity prior to InSite Remote Service Platform (RSvP) upgrade on page 25 to confirm the customer's Internet connection before the software is upgraded to RSvP.

2.7.2 Checking Internet connectivity prior to InSite Remote Service Platform (RSvP) upgrade

Makes sure an InSite1 connected system has Internet connectivity prior to a Remote Service Platform (RSvP) upgrade.

1. **(For Customers connected through DNS)** Configure and test DNS using the following instructions:
    a. Configure DNS:
        - The MR software supports DNS configuration. Check the ***USE DNS*** box and enter the DNS IP addresses on the ***Network Info*** tab of ***Guided Install***.

Figure 17 Network info DNS entry



- The MR software does not support DNS configuration. Do the following to configure DNS:

  i. Launch ***Guided Install***

  ii. Make sure the ***Network Info*** tab has correct network configuration information for ***IP/Netmask/Gateway***.

  iii. Click on ***File*** > ***quit***.

  iv. Open a terminal and login as root.

  v. Type **`gedit /etc/nsswitch.conf`** and press ***Enter***.

  vi. Add **`dns`** to the entry at the end of the file: `hosts: files dns`.

  vii. Type **`gedit /etc/resolv.conf`** and press ***Enter***.

  viii. Add Google or any other DNS entries:

    - `nameserver <site DNS IP address 1>`
    - `nameserver <site DNS IP address 2>`

  ix. Save and quit the edit terminal.

  x. Restart the system.

b. Test connectivity using DNS:

   i. Execute the following command in a terminal: **>> openssl s_client -connect insite.gehealthcare.com:443**.

      NOTE

      If the system has connectivity, the digital certificate information will be displayed and the cursor will be waiting for the next command.

   ii. Enter the following at the prompt exactly as written, including the spaces: **GET / HTTP/1.0**

   iii. The system will respond with:

```
HTTP/1.0 200 OK

Server: BigIP

Connection: close

Content-Length: 148

<HTML><HEAD><TITLE>GE Healthcare Web Server</TITLE></
HEAD><BODY><P><A HREF="http://gehealthcare.com/">GE
Healthcare</A> Web Server</P></BODY></HTML>read:errno=0
```

   iv. If the test fails, use a public IP to determine if DNS resolution is the cause of the failed test. Type the following command and then repeat 2: **>> openssl s_client -connect 198.169.189.25:443.**

2. **(For Customers connected through a proxy)** Execute the following command in a terminal:

   - **(For Proxy without login) >> env https_proxy=<proxy IP>:<port #> curl -k https://198.169.189.25:443**
   - **(For Proxy with login) >> env https_proxy=http://<user>:<password>@<proxy IP>:<port #> curl -k https://198.169.189.25:443**

   The system should respond with:

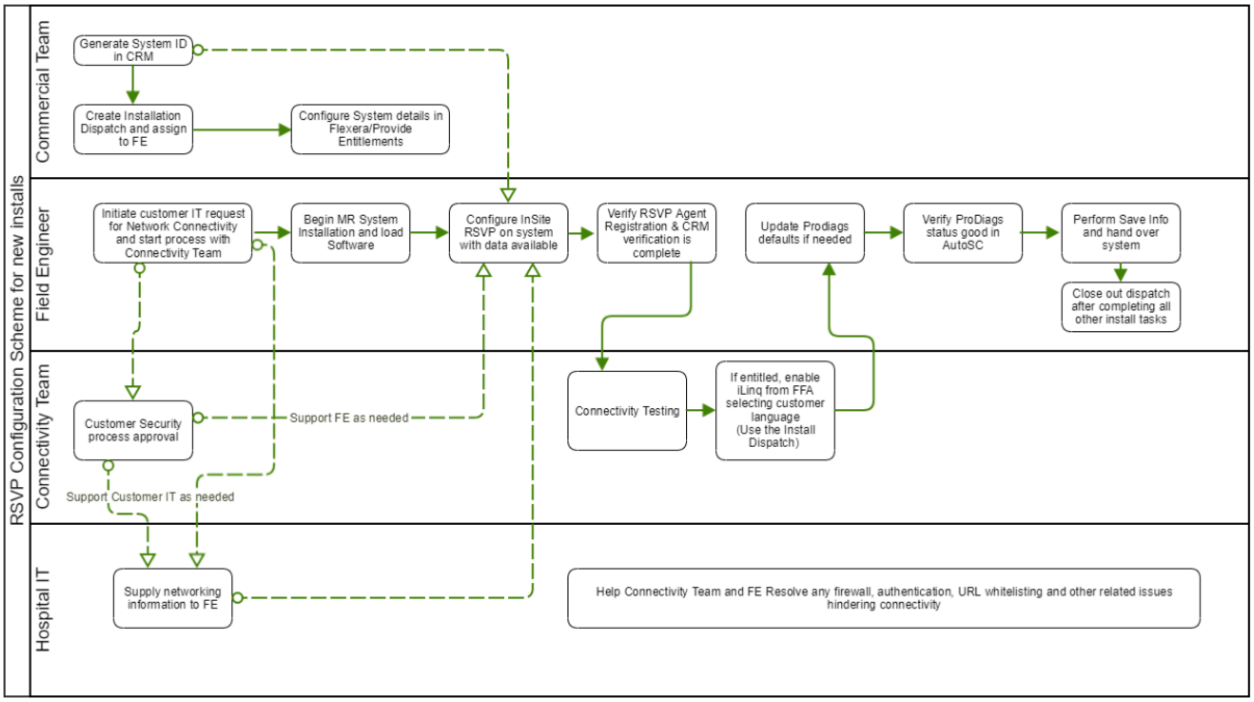
```
<HTML><HEAD><TITLE>GE Healthcare Web Server</TITLE></
HEAD><BODY><P><A HREF="http://gehealthcare.com/">GE Healthcare</A>
Web Server</P></BODY></HTML>read:errno=0
```

2.7.3 InSite Remote Service Platform (RSvP) registration process flow for forward production installations

Describes the registration process for InSite Remote Service Platform (RSvP) for forward productions installations.

The workflow shown below is a tentative representation of the connectivity process for new MR system installs. Please follow your regional connectivity process as laid out by the connectivity team.

Figure 18 Forward production registration process



2.7.4 InSite Remote Service Platform (RSvP) registration process flow for upgrades

Describes the registration process for InSite Remote Service Platform (RSvP) for upgrades.

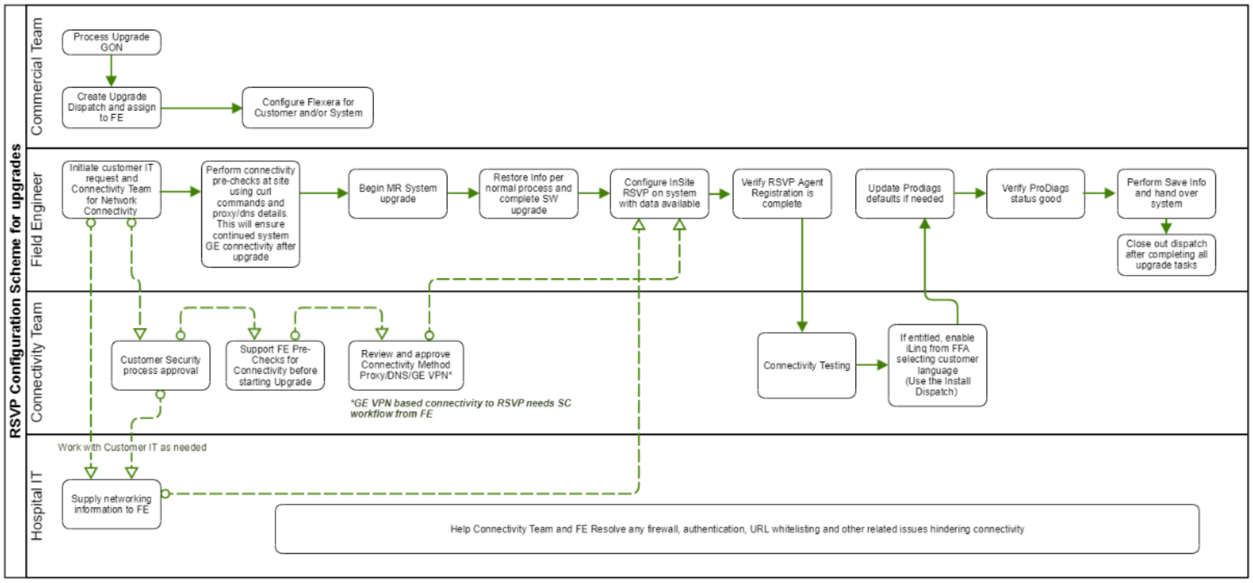
- Current InSite IIP sites have VPN routers at the site.
- This router establishes a VPN connection to the GE back office.
- On systems upgrading from InSite1, RSvP will connect through the same router directly to the RSvP server based on the URL provided during configuration.
- If the customer has concerns connecting directly to internet, the issue must be escalated through regional processes to the local connectivity team who could consider configuring the GE InSite proxy IP address.

NOTE

Configuring the system to connect through the GE VPN will slow down the connection and delay benefits of remote utilities to the customer along with disabling remote software updates to the system.

The workflow shown below is a tentative representation of the connectivity process for MR system upgrades. Please follow your regional connectivity process as laid out by the connectivity team.

Figure 19 Upgrade registration process



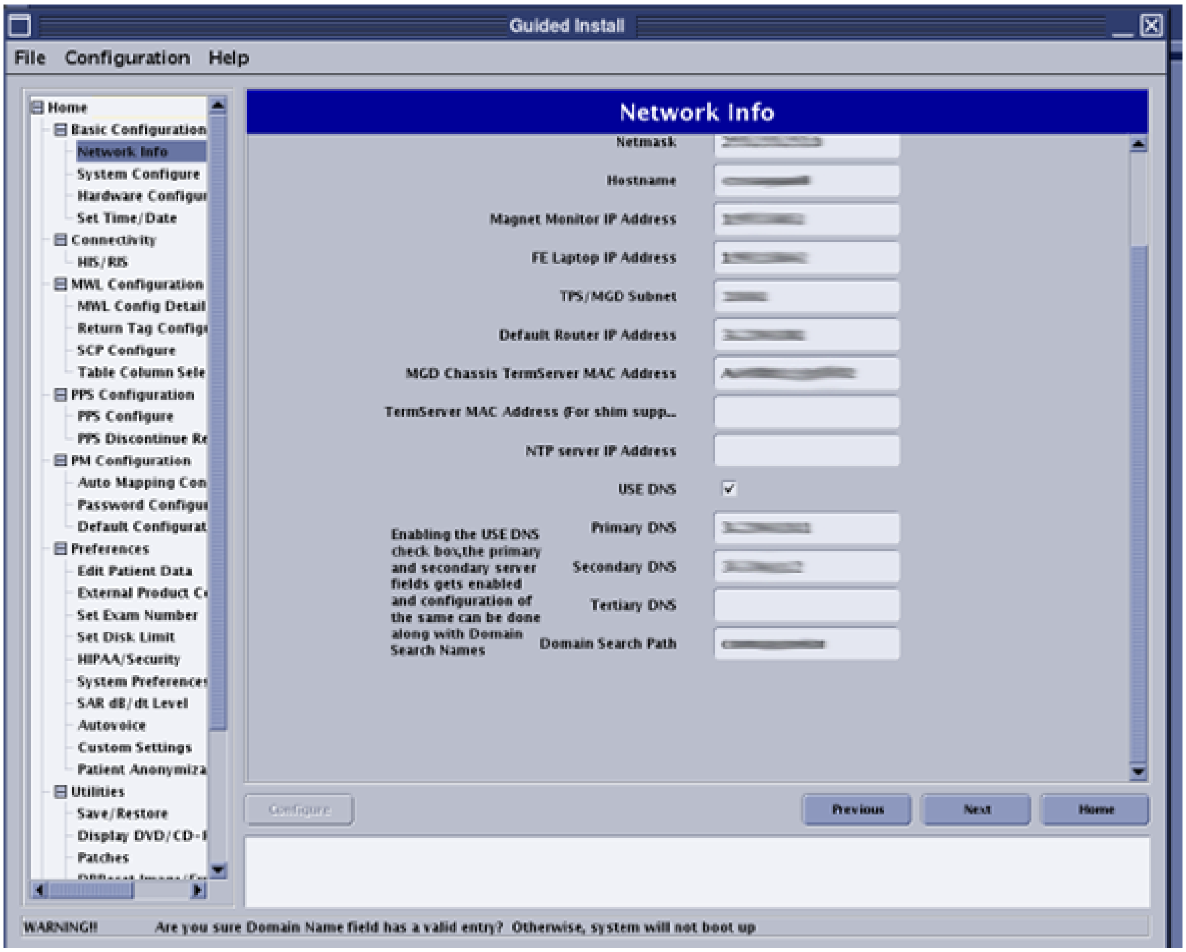
2.7.5 Configuring the Remote Service Platform (RSvP)

Configures Remote Service Platform (RSvP) for operational use.

For mobile sites, do Configuring the Remote Service Platform (RSvP) for a mobile site on page 41.

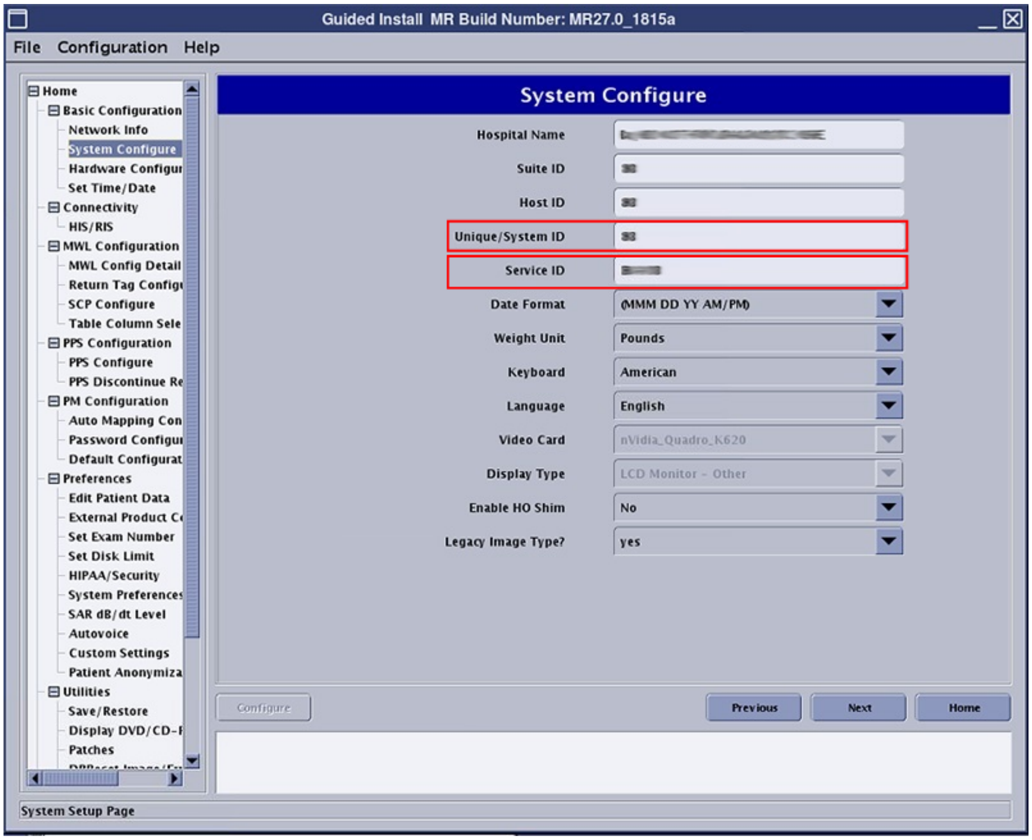
1. Do Accessing Guided Install on page 39.
2. If applicable, enable and configure DNS information in the ***Network Info*** tab.

Figure 20 Guided install network info



3. Configure the ***System ID*** and ***Service ID*** in the ***System Configure*** tab of ***Guided Install*** to the same system ID. Both fields are used interchangeably across various remote features and should be configured to be the same as the system ID of the MR system.

Figure 21 System ID and Service ID

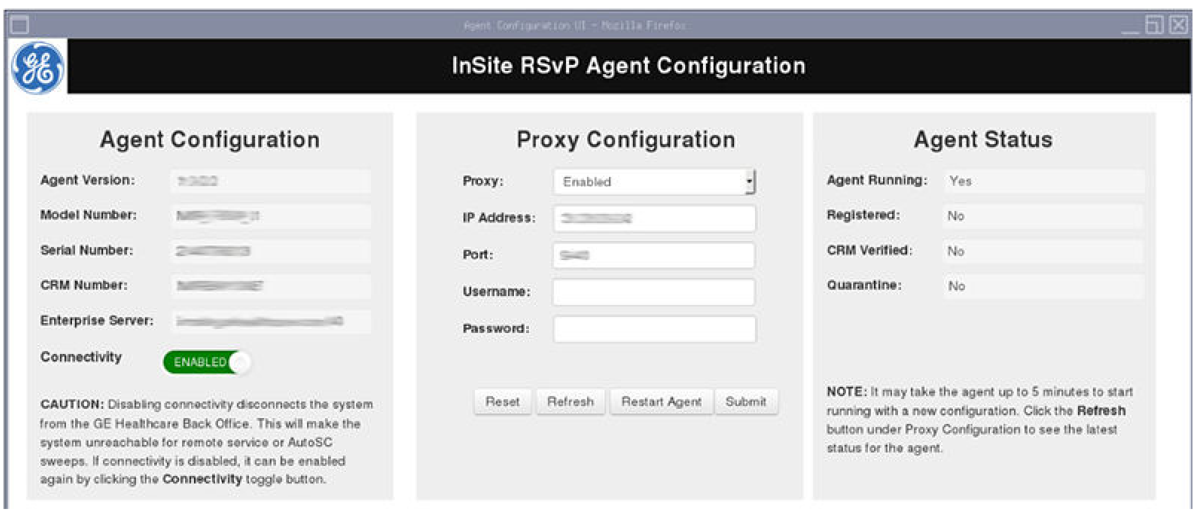


4. Navigate to the ***Service SW Insite*** window and click ***Configure Insite/IIP***.

Figure 22 Service SW Insite

5. If the site is connecting through a proxy server, enable proxy configuration and provide the IP address/proxy information from the site.

Figure 23 Proxy configuration



NOTE

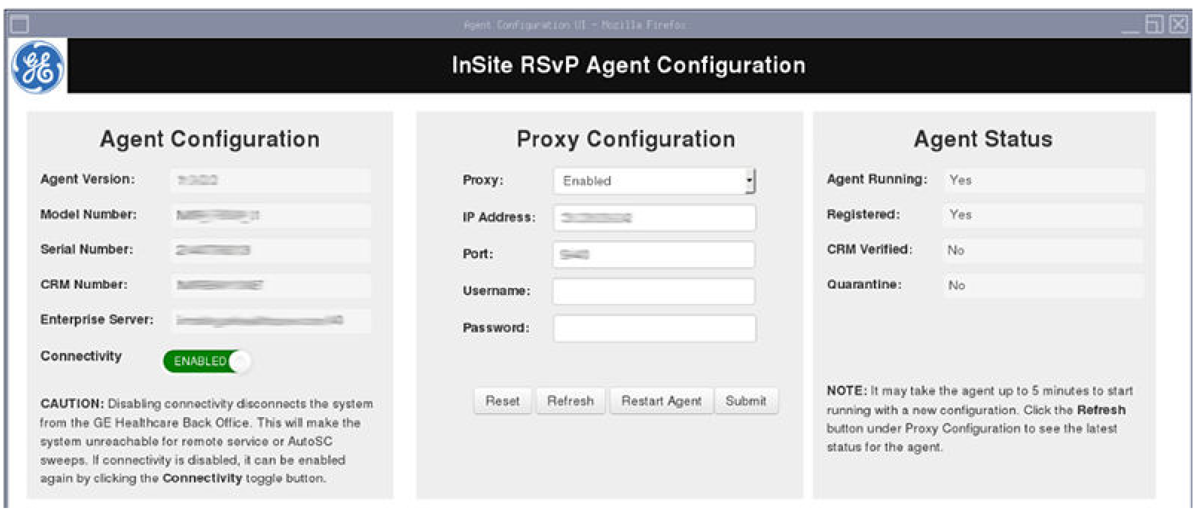
If a customer insists on using GE InSite Proxy server IP address, then obtain approval for this form of connectivity from the local connectivity team and configure the GE Proxy IP in this UI. This Proxy Server IP address is typically 150.2.1.251 with Port: 8002. Confirm this IP address and port# with your local connectivity team.

NOTE

In certain sites, the site VPN router could be on a different subnet as that of the default gateway at site. In such cases, the MR system cannot directly connect to the GE Proxy. A route should be added in the system network configuration to enable this connection. This route does not carry over across software loads through restore info; therefore, this procedure needs to be repeated in case of a software reload. See **Creating Routing Table for InSite RSvP connectivity through GE VPN Proxy** on page 42.

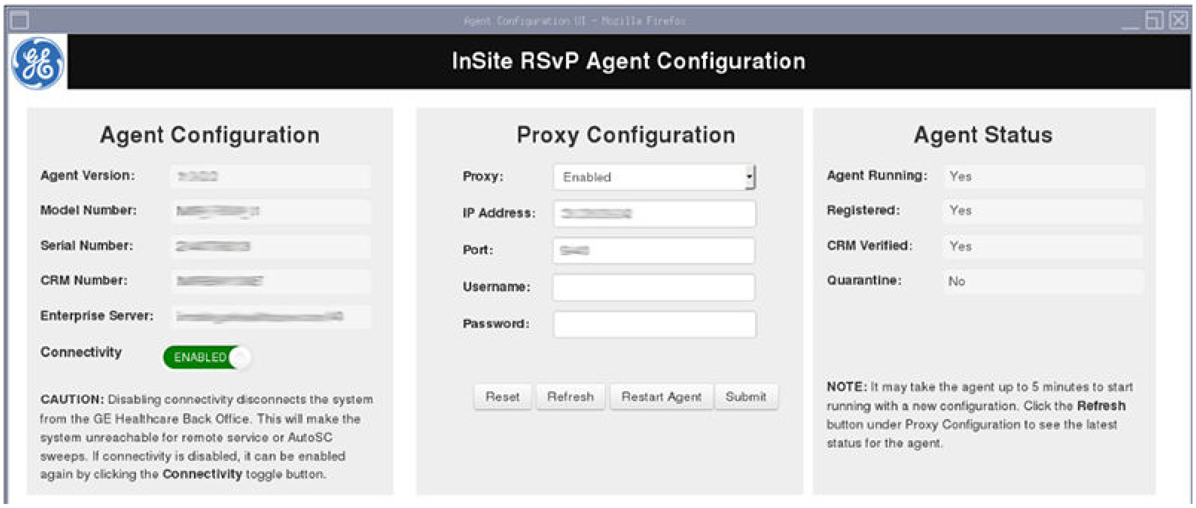
6. Click ***Submit***. A few minutes later (approximately 4 minutes) `Registered` will change to `Yes`. Click ***Refresh*** multiple times as needed.

Figure 24 Registered



7. A few minutes later (approximately 4 minutes) `CRM Verified` will change to `Yes`. Click ***Refresh*** multiple times as needed.

Figure 25 CRM verified



8. The InSite RSvP registration is complete.

2.7.6 Enabling the Remote Service Platform (RSvP) iLinq

Configures and enables iLinq through the Remote Service Platform (RSvP).

Required conditions
FFA access is required to enable iLinq on the system. If you do not have access to FFA and RSvP, contact your regional connectivity team or the MR support team to enable iLinq.
RSvP registration must be completed on the system with `Registered` and `CRM Verified` set to `Yes`.

NOTE

It may take up to 10 minutes after RSvP registration is completed for the system ID to display in FFA.

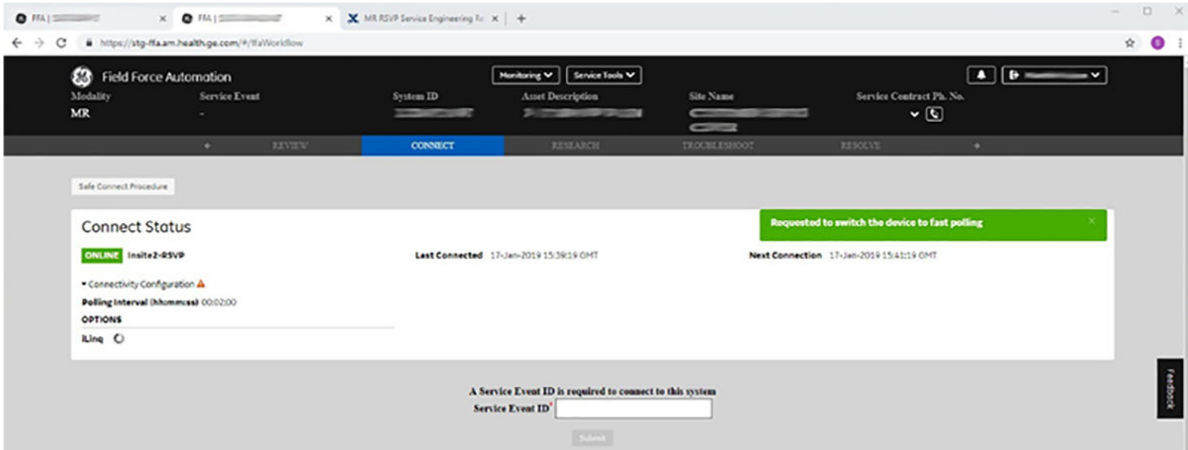
NOTE

The iLinq page will be blank until iLinq is enabled from FFA.

1. Create a service record for the site with its system ID and launch FFA.
2. Navigate to the ***CONNECT*** tab.

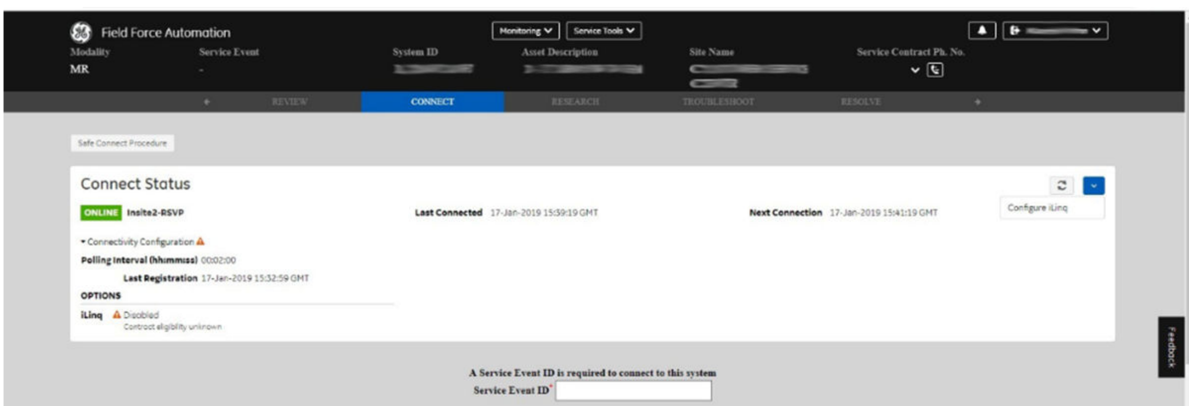
3. Make sure `Connect Status` for the system shows `ONLINE` and ***Connectivity Type*** is set to ***Insite2-RSvP***.

Figure 26 iLinq connect status



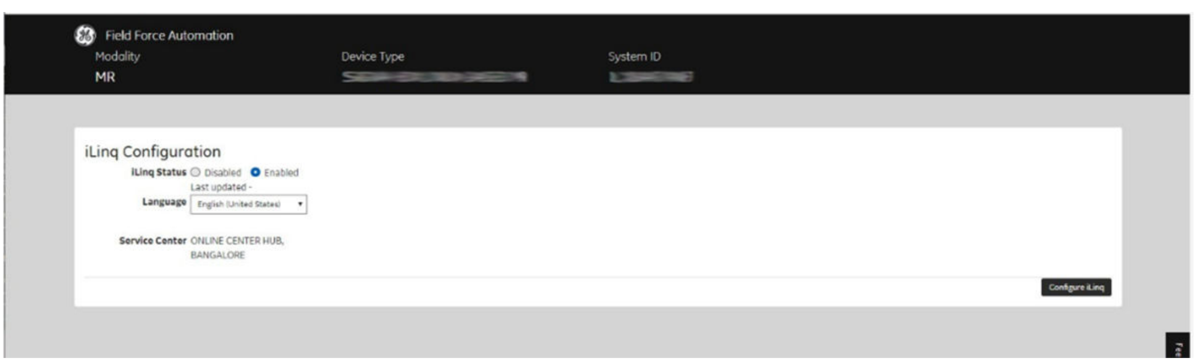
4. From the menu, select ***Configure iLinq***.

Figure 27 Configure iLinq



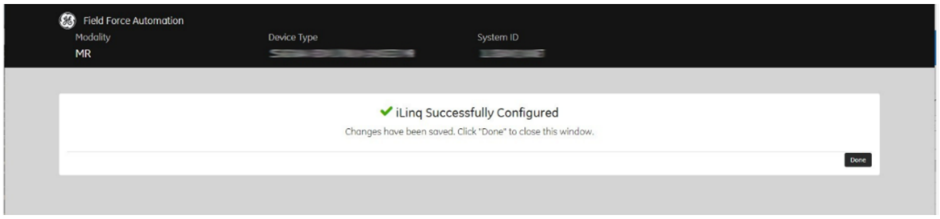
5. Set ***iLinq Status*** to ***Enabled***. Select the correct language for the site from the drop-down list, and click ***Configure iLinq***.

Figure 28 iLinq configuration



6. Click ***Done*** to close the window.

Figure 29 iLinq configuration successful

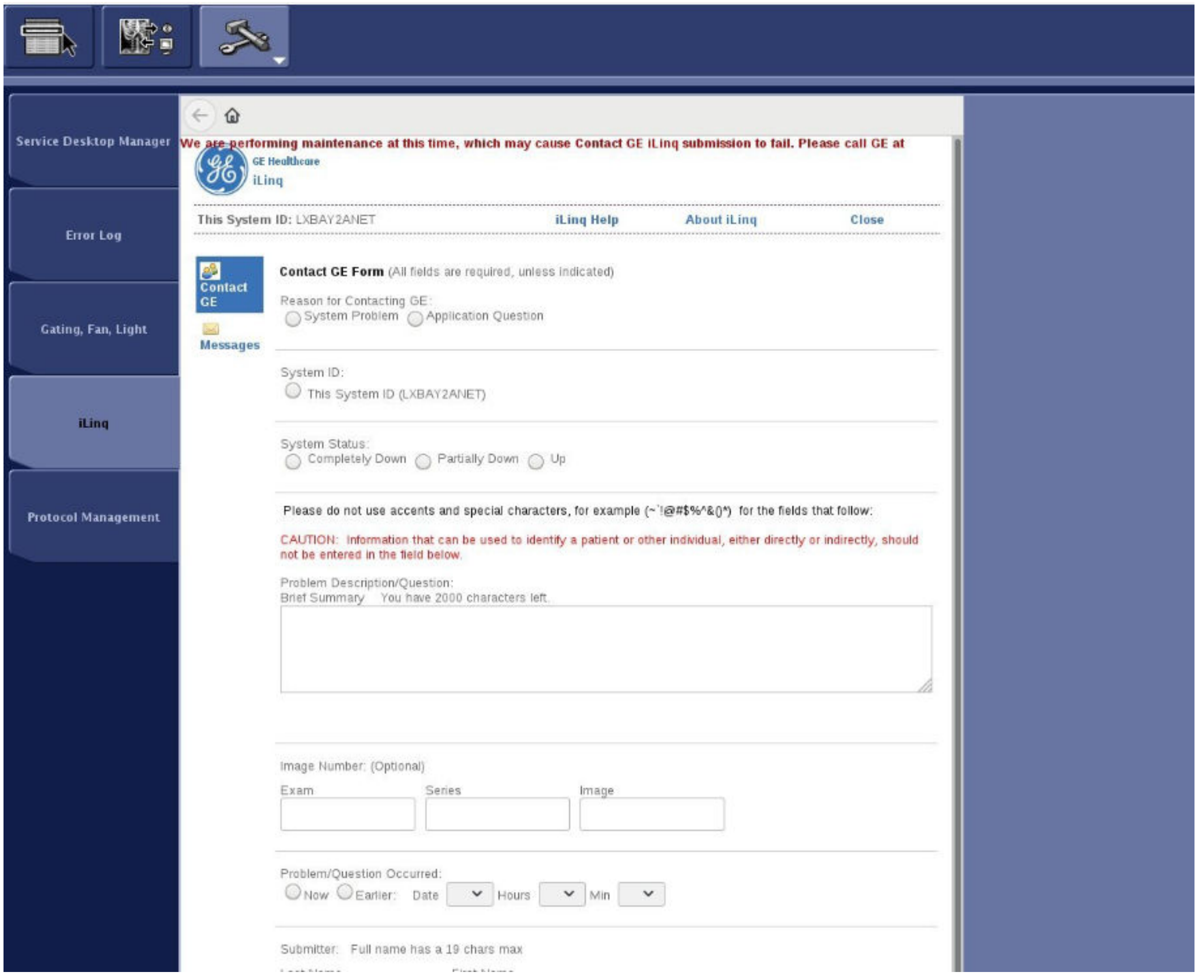


7. After the iLinq checkout has been completed, reboot the host PC. The iLinq button will display as shown below.

Figure 30 iLinq button

8. Go to the ***iLinq*** tab on the service desktop. The iLinq page should display. If the iLinq page does not display, click on the ***iLinq*** icon. This may take up to 15 seconds on the first attempt.

Figure 31 iLinq page



NOTE

Since iLinq is always enabled remotely, a SaveInfo is often not done to immediately save this setting. This results in the iLinq feature being unavailable after a software reload. Therefore, make sure a SaveInfo is done immediately after enabling iLinq from FFA.

2.7.7 InSite Remote Service Platform (RSvP) default site settings

Default site settings for InSite Remote Service Platform (RSvP).

The following table contains the default software implementation. This can be changed by the customer as required if there are any concerns.

Table 4 Default software implementation

Site	SW Content	Connectivity		Sweeps and ProDiags		Remote Service (iLinq, CSD, SSH, SFTP)		Flexera Download		Agent Status on Device
		Default	Options	Default	Options	Default	Options	Default	Options	
Contract/N on Contract	YES	ON	YES/NO	ON	YES/NO	ON	YES/NO	ON	YES/NO	ON

Figure 32 Status of data flow to AutoSc, software download, and remote service features

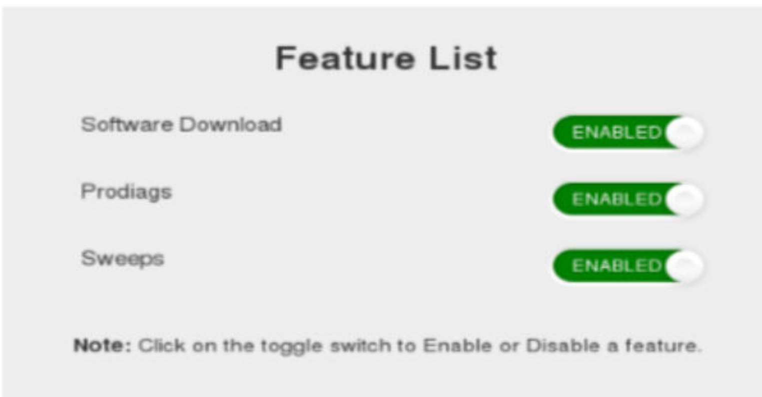


Figure 33 Status of remote connectivity

Accessing Guided Install

This procedure describes how to invoke Guided Install as root or open Guided Install in FE mode.

Required conditions
A Class M SSA key or Class C/A2 SSA soft key license is needed to access Guided Install from SIGNA applications. It is not needed when launched as Root. Refer to *Service licensing* for information on how to install an SSA soft license.

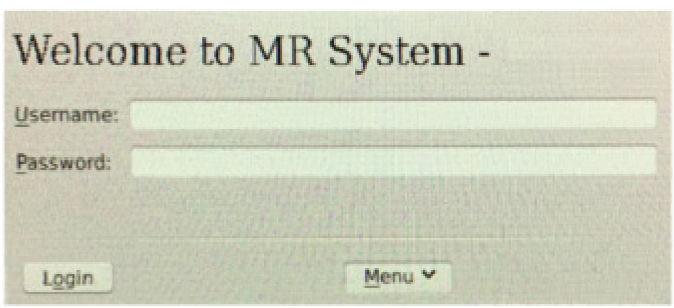
1. Choose one of the following methods to access Guided Install.
    - Invoke Guided Install as Root.
    - Open Guided Install in FE Mode.
2. Invoke Guided Install as root.

a. When you power up the host computer and see the login screen, click on ***root login***.

Figure 34 Login screen

b. Log in with the user name: `root` and password: `operator`

Figure 35 Welcome login screen



NOTE

It is possible that the customer changed the default password. If you cannot log in, contact the customer for the correct password.

Logging in as the root user automatically displays the Linux desktop and launches the ***Guided Install Starter*** window.

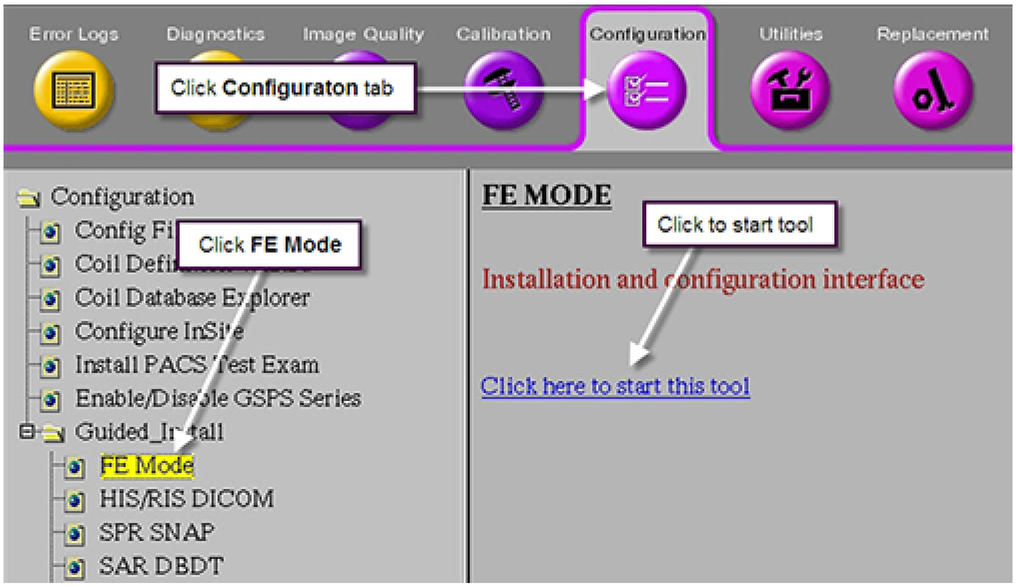
c. Click ***Yes*** to invoke the guided install.

NOTE

If ***No*** has already been selected, right-click on the Linux desktop and select ***Start Guided Install***.

3. Open Guided Install in FE Mode.
   a. Insert the Service Key.
   b. Open the Common Service Desktop.
   c. Under the ***Configuration*** tab, select ***Guided Install*** > ***FE Mode*** > ***Click here to start this tool***.

   Figure 36 Guided Install FE Mode

   d. When the C shell opens, type the root password.

   NOTE

   The default root password is `operator`. It is possible that the customer changed the password. If you cannot log in, contact the customer for the correct password.

Configuring the Remote Service Platform (RSvP) for a mobile site

Additional steps required for configuring the Remote Service Platform (RSvP) on a mobile site.

The configuration will need to be repeated at every location of the mobile system. SaveInfo and RestoreInfo will carry over and automatically restore connectivity during system reboot.

1. Update the Sub or Child System ID in ***Guided Install*** under the ***System ID*** and ***Service ID*** field in ***Guided Install***.
2. Follow the configuration procedure in 2.7.5 Configuring the Remote Service Platform (RSvP) on page 29 and submit for RSvP registration.

Creating Routing Table for InSite RSvP connectivity through GE VPN Proxy

In certain customer sites where the VPN router IP Address is not in the same subnet as that of the customer site gateway IP address, a route is needed in order for these MR systems to connect to the InSite back office.

2.8.1 Adding a Static Route

To add a persistent route to the external interface of the MR scanner, follow these steps:

1. Determine the parameters needed to set up the static route:
   - the ***network address*** of the destination network: 150.2.0.0
   - the ***netmask*** to determine which hosts are in and out of the destination network: 255.255.0.0
   - the ***gateway*** address to send these packets to: This detail needs to be obtained from the customer IT department
2. Open a command prompt and change over to root user using the "su –" command
3. Determine the name of the external interface of the system using the ***mr-network-cli*** command

   ```
   [root@t19 ~]# /usr/local/bin/mr-network-cli --get-interface-name external

   enp0s25
   ```

   The output of the above command indicates the name of the external interface is ***enp0s25***. Replace <ext eth port> in the steps below with the name of the interface for your specific machine.
4. Create the ***/etc/sysconfig/network-scripts/route-<ext eth port>*** file as root

   ```
   [root@t19 ~]# gedit /etc/sysconfig/network-scripts/route-<ext eth port>
   ```
5. Populate it with the following information, then save and quit:

   ```
   ADDRESS0=150.2.0.0
   NETMASK0=255.255.0.0
   GATEWAY0=x.x.x.x
   ```

   where x.x.x.x is the customer site gateway ip address
6. Restart the networking subsystem with the ***service*** command:

   ```
   [root@t19 ~]# service network restart
   Restarting network (via systemctl): [ OK ]
   root@t19 ~]#
   ```
7. Confirm that the route was added - use the ***ip route*** command to check:
8.
   ```
   [root@t19 ~]# ip route
   ...
   ```

```
150.2.0.0/16 via x.x.x.x dev enp0s25
...
```

The static route will be added at boot time when the interface is brought up.

You can add more such static routes as needed in the route-<ext eth port> file using incremental suffixes for ADDRESS, NETMASK, and GATEWAY for the additional entries. Example:

- the second static route would use ADDRESS1, NETMASK1, GATEWAY1
- the third static route would use ADDRESS2, NETMASK2, GATEWAY2

2.8.2 Removing the Static Route

Remove the **/etc/sysconfig/network-scripts/route-<ext eth port>** file to remove all external static routes. You can remove the **ADDRESSn**, **NETMASKn**, and **GATEWAYn** entries from the **same** file if you want to remove just a single route. Restart the network with **service network restart** to remove the route on the running machine without rebooting.

The network configuration scripts cannot handle gaps between entries, so make sure any remaining entries have contiguous numbering. Given the following example:

```
ADDRESS0=...
NETMASK0=...
GATEWAY0=...
ADDRESS1=...
NETMASK1=...
GATEWAY1=...
ADDRESS2=...
NETMASK2=...
GATEWAY2=...
```

Removing ADDRESS2, NETMASK2, and GATEWAY2 without renumbering is okay.

Removing ADDRESS1, NETMASK1, and GATEWAY1 without renumbering ADDRESS2, NETMASK2, and GATEWAY2 will result in this route not being applied.

Page intentionally left blank

3 Appendix

3.1 Running the About MR Scanner tool

The ***About MR Scanner*** tool displays scanner information. This procedure confirms that this information is correct.

Personnel requirements			
Required persons	**Preliminary requirements**	**Procedure**	**Finalization**
1	-	5 minutes	-

Required conditions
Complete software setup before running the ***About MR Scanner*** tool.
Note down the serial number on the magnet rating plate that is physically located on the scanner.
Review the magnet and system configurations in Guided Install and take note of the values before running ***About MR Scanner***.
A service key is required to launch the ***Installation and Calibration Wizard***.

The ***About MR Scanner*** tool displays information and provides values for:

- Magnet serial number
- Magnet enclosure type
- Field strength
- Gradient type

This procedure is to confirm that the scanner information displayed in the ***About MR Scanner*** tool is correct.

The ***About MR Scanner*** tool displays values recommended by International Electrotechnical Commission (IEC) standards. The hardware configuration and product configuration values selected while executing this tool determines the safety information to be shown specifically for the site system. Safety data will not be displayed until this procedure has been completed. If any of the configurations are modified, the safety data will no longer be displayed until this procedure is repeated. This feature does not impact functionality of any other subsystem.

1. Verify the serial number on the magnet rating plate matches the value listed on the ***Guided Install Hardware*** configuration screen. See Accessing Guided Install on page 39.
2. Launch the ***About MR Scanner*** tool.
   - From the Common Service Desktop select ***Utilities*** > ***About MR Scanner***.

An alert message will require verification that setup steps have been completed before activating this tool.

3. Click ***OK*** to continue.

   A configuration message displays asking you to verify the magnet rating plate.

4. Verify that the magnet serial number matches what is shown in the tool.

   - If the magnet rating plate serial number matches, type **`Yes`** and press ***Enter***.

     When prompted, enter the number associated with the system product name.

     The ***About MR Scanner*** window will open and display system information.

   - If the magnet rating plate serial number does not match, type **`No`** and press ***Enter***.

     When prompted, use the ***Guided Install*** to confirm hardware configuration before running this tool again.

     Update the hardware configuration as needed. See System and hardware configuration.

5. Manually check the ***About MR Scanner*** window and verify that the displayed scanner and the system information matches the site. Click ***Close***.

   A message displays requesting confirmation that the system configuration is correct.

6. Confirm system configuration.

   - Click ***Yes*** if all the details are correct.

     The tool is complete.

   - Click ***No*** if the information is incorrect.

     Modify the configuration values from the ***Guided Install*** menu and then relaunch the ***About MR Scanner*** tool. See System and hardware configuration.

3.2 Running LVShim Gradshim

The objective is to complete fine tuning of the homogeneity of the magnet.

Safety
Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare *MR Service Safety Manual* (5452735). • Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

The Gradshim process is done with a sampling diameter of 22 cm DSV and the same bandwidth as LVShim Fine, which depends on the quality (good/bad) of the fine tuning. Once the scan passes and all harmonics are within specification, the magnetic field is reanalyzed in Test mode automatically at 45 DSV for customer acceptance.

The gradshim process includes LVShim - test. The objective of this test is to analyze completed scans at 45 DSV without calculating currents.

By default, gradshim is performed in auto mode, although manual mode is also available.

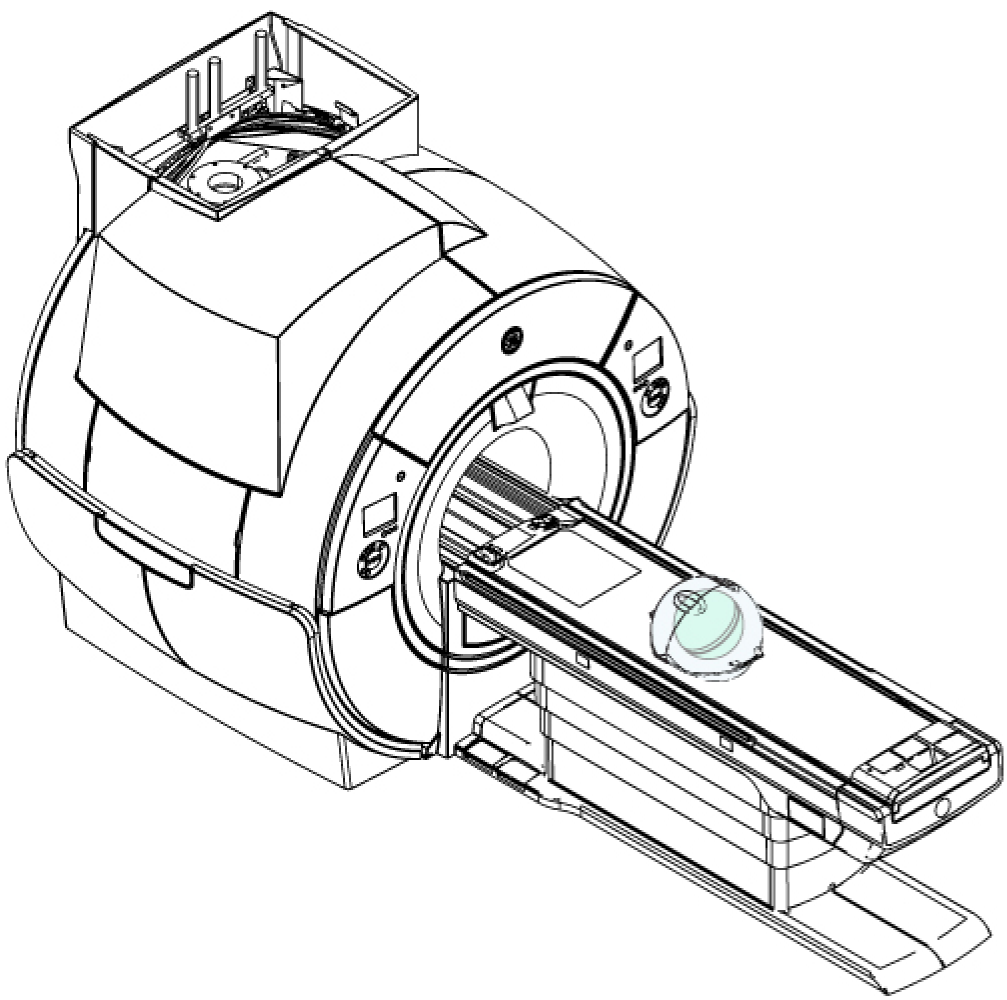
NOTICE

PERFORM A LOCALIZER SCAN TO VERIFY THAT THE PHANTOM IS CENTERED BEFORE PERFORMING THE LVSHIM.

To set up the phantoms and system for this procedure, follow the instructions and procedures.

1. Make sure that all coils have been removed from the cradle and phantoms from the bore.
2. Place the LVShim phantom on the patient table.

Figure 37 Placing the LVShim phantom on the patient table



3. Set up and align the LVShim phantom. Align the phantom so that:
    a. The phantom's seam is oriented perpendicular to the table.
    b. The phantom's z-axis is along the length of the cradle.
    c. The phantom is centered left to right with respect to the cradle top.
4. Landmark on the LVShim phantom crosshairs.
5. Press ***Advance to Scan***.

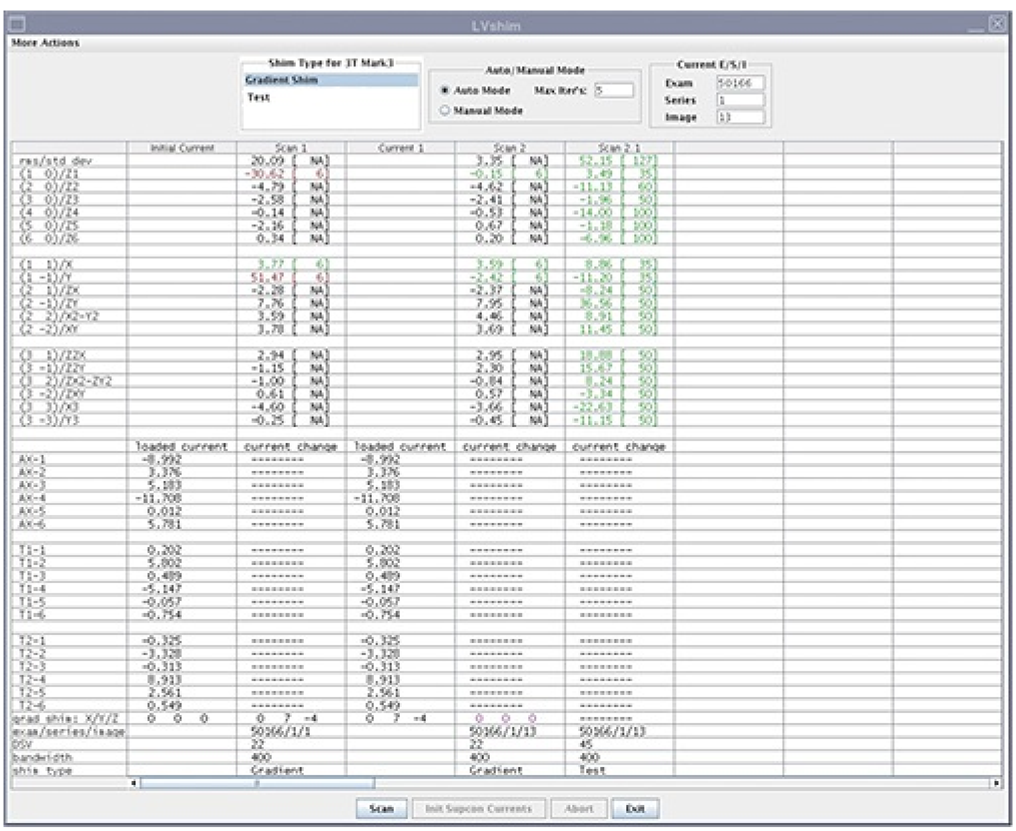
3.2.1 Running LVShim in Auto mode

In auto mode, gradient currents are automatically calculated after you click ***Scan*** to fine tune the homogeneity of the magnet. In this mode, the tool stops after the X, Y and Z harmonics are within specification. In test mode, the image from the last scan is automatically reanalyzed.

1. Start the LVShim tool:
    - (For non-proprietary mode) From the Common Service Desktop, select ***Calibration*** > ***LVShim***, then select ***Click here to start this tool***.

    By default, Gradshim is performed in auto mode, though manual mode is also available.
2. Select ***Gradient Shim*** and click ***Scan***.

Figure 38 Gradient shim - auto mode



NOTE

For Gradshim software version PX26.2 and above , the gradient shim bit resolution was increased by 16 times. This will result in gradient current values that are much higher than those observed in previous versions of this tool.

In auto mode, the tool does the following:

- Sets up the protocol and collects image data
- Calculates linear harmonics and compares it with the specification
- Calculates the delta currents and the new currents based on the harmonics (dc offsets for X, Y, and Z)
- Completes multiple iterations (maximum = five) until harmonics meets the specification
- Shows the reanalysis of the last iteration in the final column in test mode (customer acceptance specification)

1. To finalize this procedure, remove any phantoms and/or tools used.
2. Perform a Check Scan before turning the unit over to the customer. See Doing a check scan.

3.2.2 Running LVShim in Manual mode

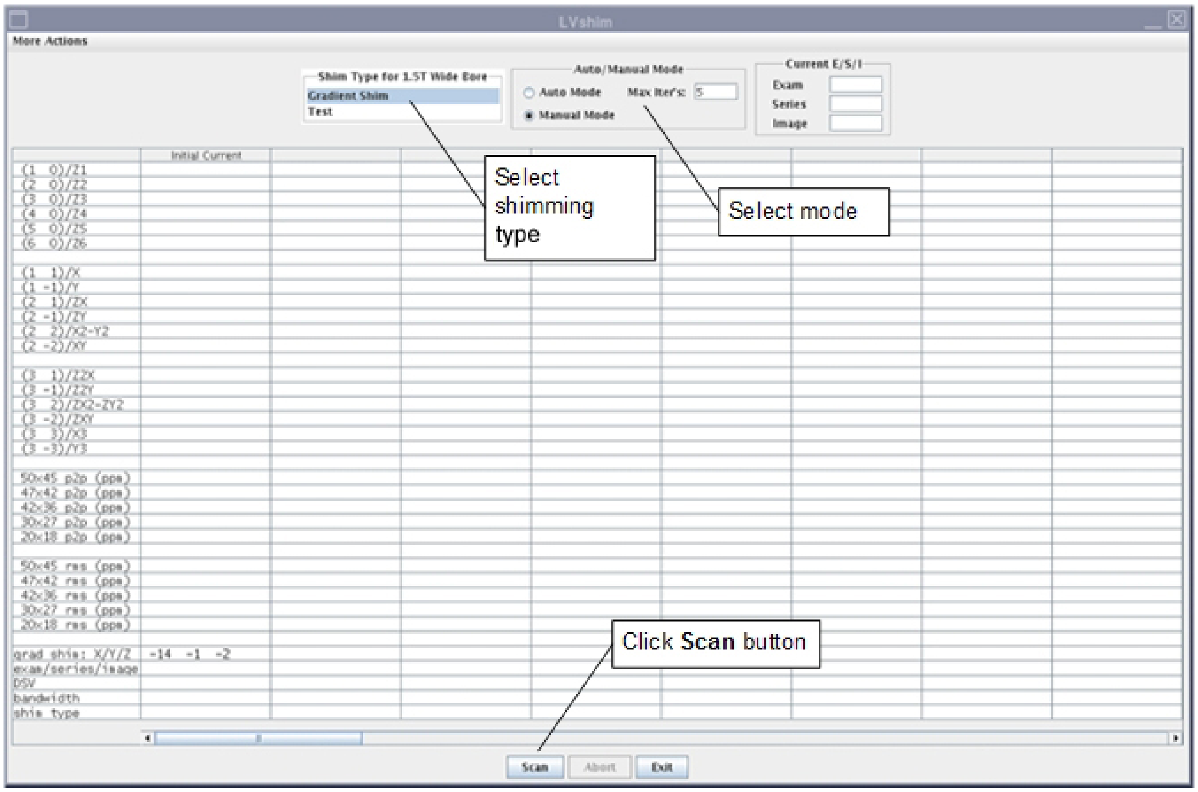
1. Start the LVShim tool:

- (For non-proprietary mode) From the Common Service Desktop, select ***Calibration*** > ***LVShim***, then select ***Click here to start this tool***.

By default, Gradshim is performed in auto mode, though manual mode is also available.

2. Select the shim type (gradient, supercon rough, supercon fine, or test) and ***Manual Mode***, then click ***Scan***.

Figure 39 LVShim - manual mode



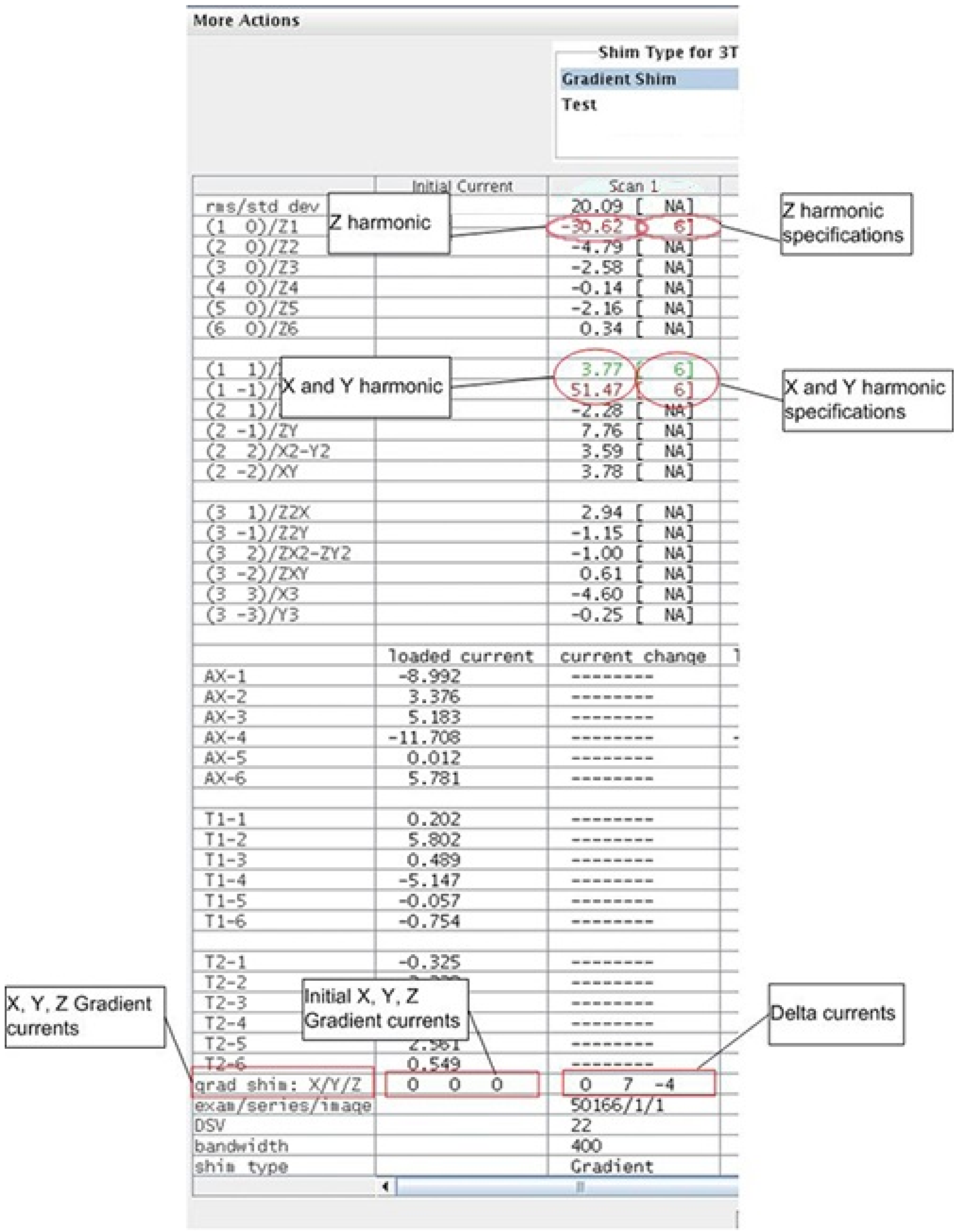
NOTE

For Gradshim software version PX26.2 and above , the gradient shim bit resolution was increased by 16 times. This will result in gradient current values that are much higher than those observed in previous versions of this tool.

After the scan is complete, the tool calculates harmonics and compares X, Y and Z to specifications.

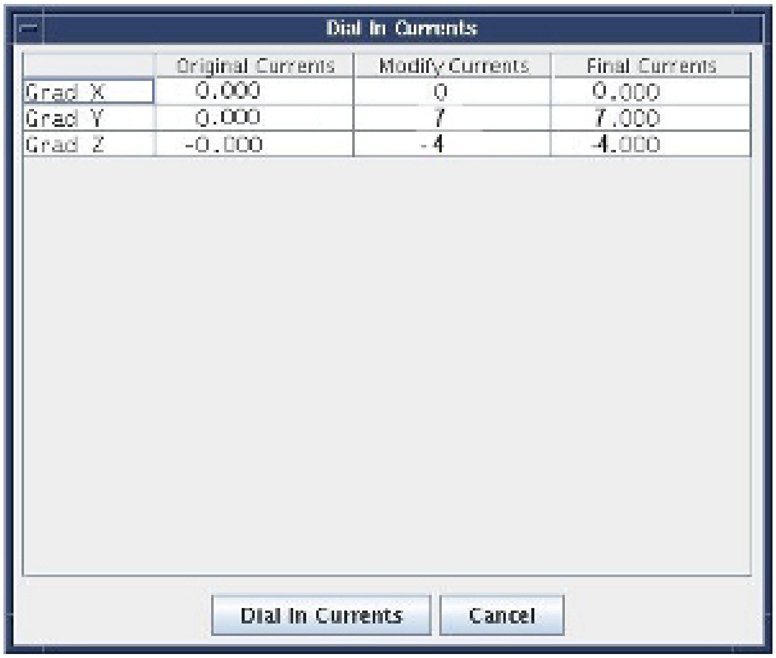
- Green cell = Passed specification
- Red cell = Failed specification

Figure 40 Gradient harmonics and gradient bits



The tool calculates the X, Y and Z gradient delta currents (for gradient bits only) and the new currents based on the harmonics in another window.

Figure 41 Original delta and new currents (examples)

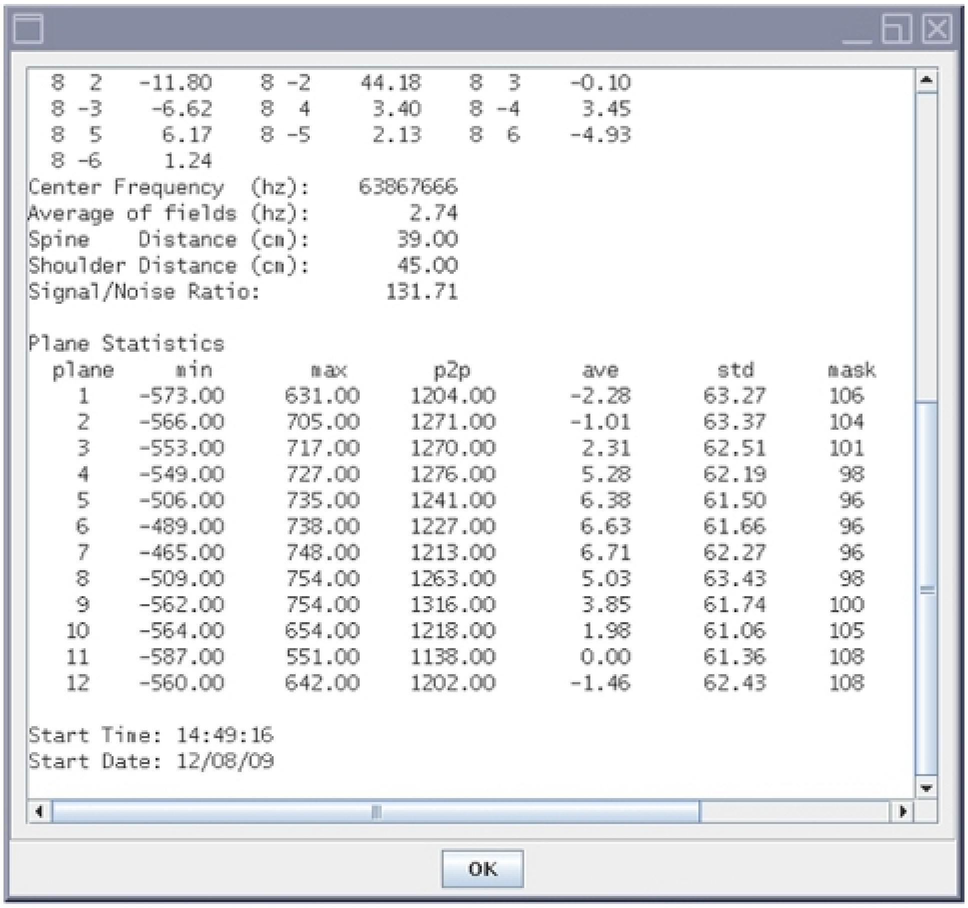


3. To accept or decline the new currents:
   - If X/Y/Z harmonics data cells are green, decline the new currents by clicking the ***Cancel*** button. Ignore cells with the specification NA.
   - If any harmonics data cells are red, accept the new currents by clicking the ***Dial In Currents*** button. The new currents will load in a new column in the LVShim user interface as shown in the *Gradient harmonics and gradient bits* figure above.

   If after five manual iterations any cell is still red, click ***More Actions*** and select ***Expert Mode***.

4. For more details about the scan (cal file name, center frequency, etc.), select ***More Actions*** > ***More Info***.

   Figure 42 More information on scan



5. When all iterations are complete and all harmonics are within specification, click ***Exit*** in the ***LVShim*** window.

   If this also fails, run passive shim using the instructions in the applicable *Magnet and Cryogens Subsystem Manual*.

1. To finalize this procedure, remove any phantoms and/or tools used.
2. Perform a Check Scan before turning the unit over to the customer. See Doing a check scan.

3.3 Doing the System Performance Test (SPT) for VX28.0

The system performance test (SPT) automatically executes Eddy Current Check, SNR, Stability and Coherent Noise tests. To do SPT in PM mode, you must launch SPT from the PM Assist menu.

Personnel requirements			
Required persons	**Preliminary requirements**	**Procedure**	**Finalization**
1	-	90 minutes	-

Tools and test equipment			
Item	**Quantity**	**Part number**	**Manufacturer**
	1	-	-

Required conditions
Cancel all previous exams. Click ***End Exam***.
Remove any patient positioning devices, such as patient restraints or other attached devices.

Safety
Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare *MR Service Safety Manual* (5452735). • Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

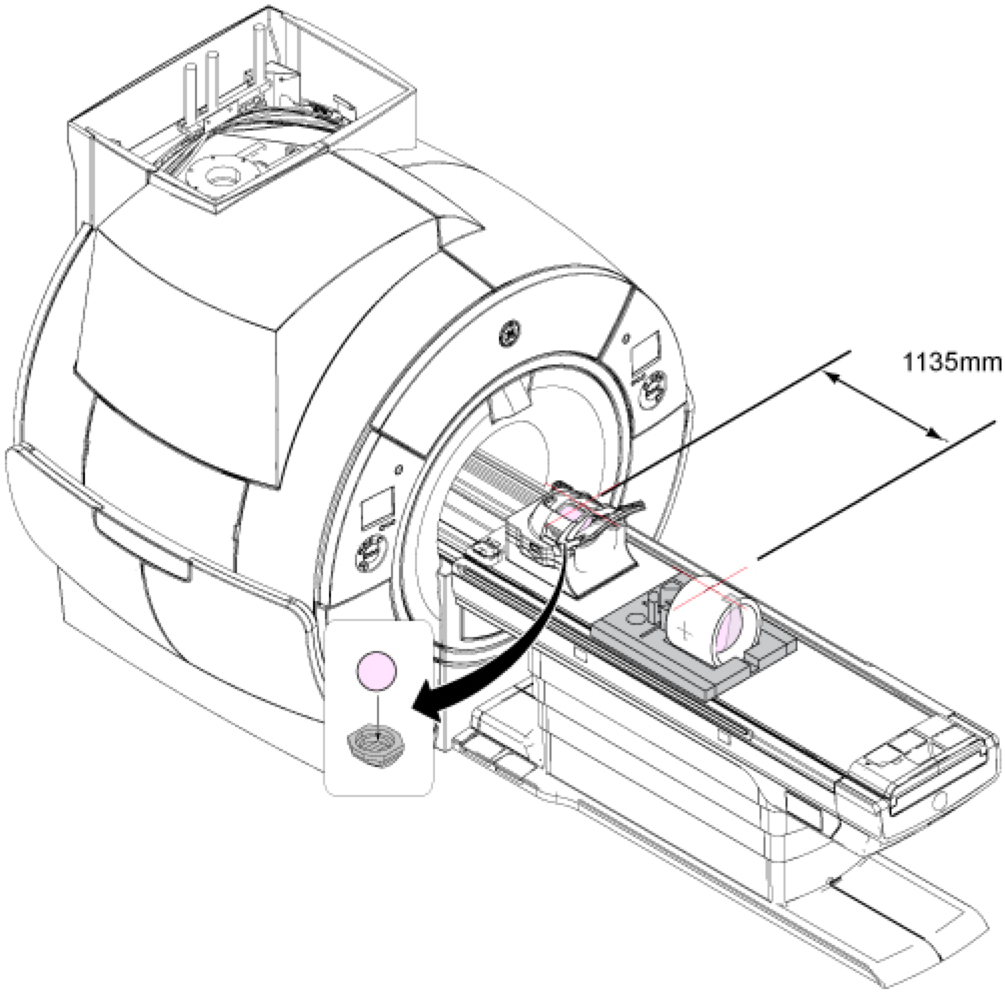
3.3.1 Doing the System Performance Test (SPT)

For this test, you set up the phantoms once and do all tests in one run.

1. Position the coil and phantoms as follows:
    a. Remove the service filler panel to uncover the cradle slot.
    b. Put the HNU coil into the cradle slot and insert the connector.
    c. Put the head TLT sphere in the HNU with the positioner.
    d. Landmark on the crosshair of the HNU, but do not advance to scan.

e. Move the cradle into the bore until the display shows `I 1135 mm`.

Figure 43 Distance between HNU landmark and center mark on body loader



f. Turn on the alignment light.

g. Put the body loader on the phantom foam support and put the body sphere in the loader.

h. Align the crosshairs on the loader to the alignment light (do not landmark).

NOTE

Do not landmark on the body sphere.

i. Press ***Advance to Scan***.

2. Launch SPT in the appropriate mode:

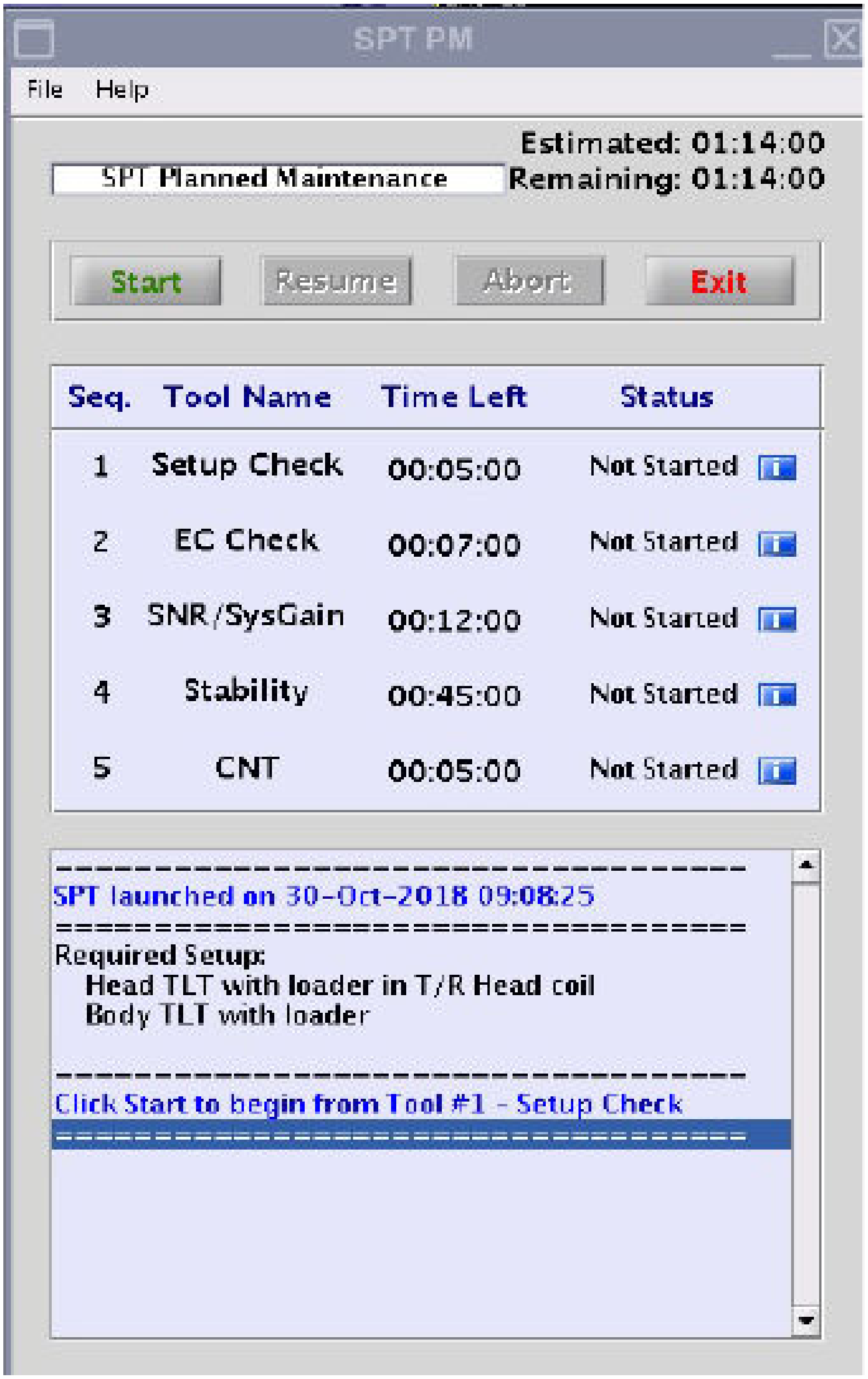
NOTE

You must run SPT in PM Mode if you are running it as part of a PM visit, otherwise, the PM check may not pass.

- If you are running SPT PM Mode, go to ***PM*** > ***PM Assist*** > ***SPT PM Mode*** > ***Click here to start this tool***.

  The ***SPT PM*** window shows.

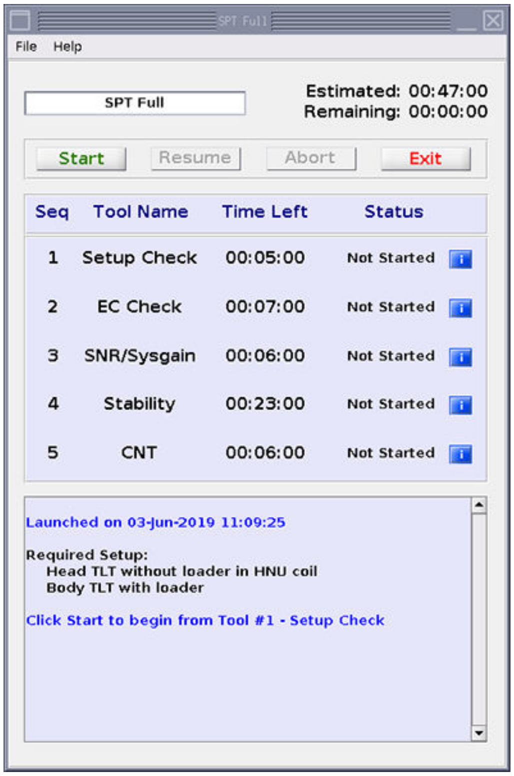
  Figure 44 SPT PM tool GUI



- To run SPT in Full Mode, go to ***Image Quality*** > ***SPT Full Test Menu*** and select ***Click here to start this tool***.

  The ***SPT Full*** window shows.

  Figure 45 SPT Full tool GUI



3. Click ***Start***.

   When SPT is complete, the message `SPT execution completed` shows in the status window.

   NOTE

   The setup check will confirm setup for both head and center locations.

4. Resolve any errors or failures and click ***Start*** to run SPT again.

   NOTE

   Do not click ***Resume*** if SPT is being called from ICW. Always click ***Start*** to complete the test from ICW.

   If you click the blue ***i*** button next to each tool (only for PASS or FAIL status), the viewer will show for that respective tool.

   After SPT has finished running the tools, a data file (for example, `YYYYMMDD_HHMMSS_<bayname>_<instance>.stc`) is created in the `/usr/g/service/data`

NOTE

Report manager is not supported for this tool. Use the Viewers to see the results in detail.

5. On the SPT interface, go to ***File*** > ***Open*** > ***Data File*** to view the overall results of the SPT run.
   See the SPT viewer to review and analyze results.
   See 3.3.2.1 System Peformance Test (SPT) user interface help on page 64 for more information.

1. If you aborted the scan, check the error log to find the cause of the problem.

3.3.2 System Performance Test (SPT) viewers and specification files

Eddy current check, SNR, stability and coherent noise tests all have viewers that show test results data. System Performance Test (SPT) Full mode is not supported by Report Manager.

The specification values, data and results can be viewed by launching the respective Viewers using the blue ***i*** button available next to each tool status.

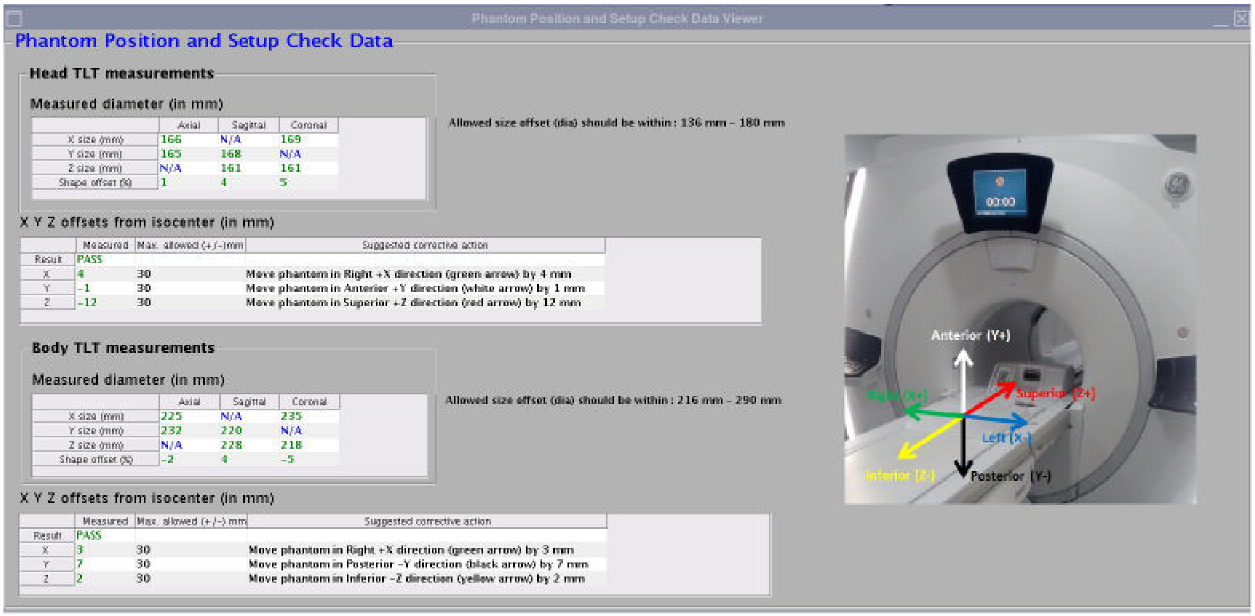
3.3.2.1 Phantom position and setup check viewer

After finishing the phantom position and setup check, a results file is created. It contains the measured results of phantom offset, shape offset, and phantom size. The latest head phantom and body phantom result files are saved under `/usr/g/service/cclass/ppsc` as `setupCheck.result_body` and `setupCheck.result_head`, respectively. The format of the two files is the same. Lines four, five, and six in those files are the detailed results from the phantom position check.

When you click the i button next to this tool on the SPT interface (only for PASS or FAIL status), the viewer shows. The viewer always shows the latest results when it launches.

The following is an example of the result file for body phantom position check (`setupCheck.result_body`):

Figure 46 Phantom position and setup check data



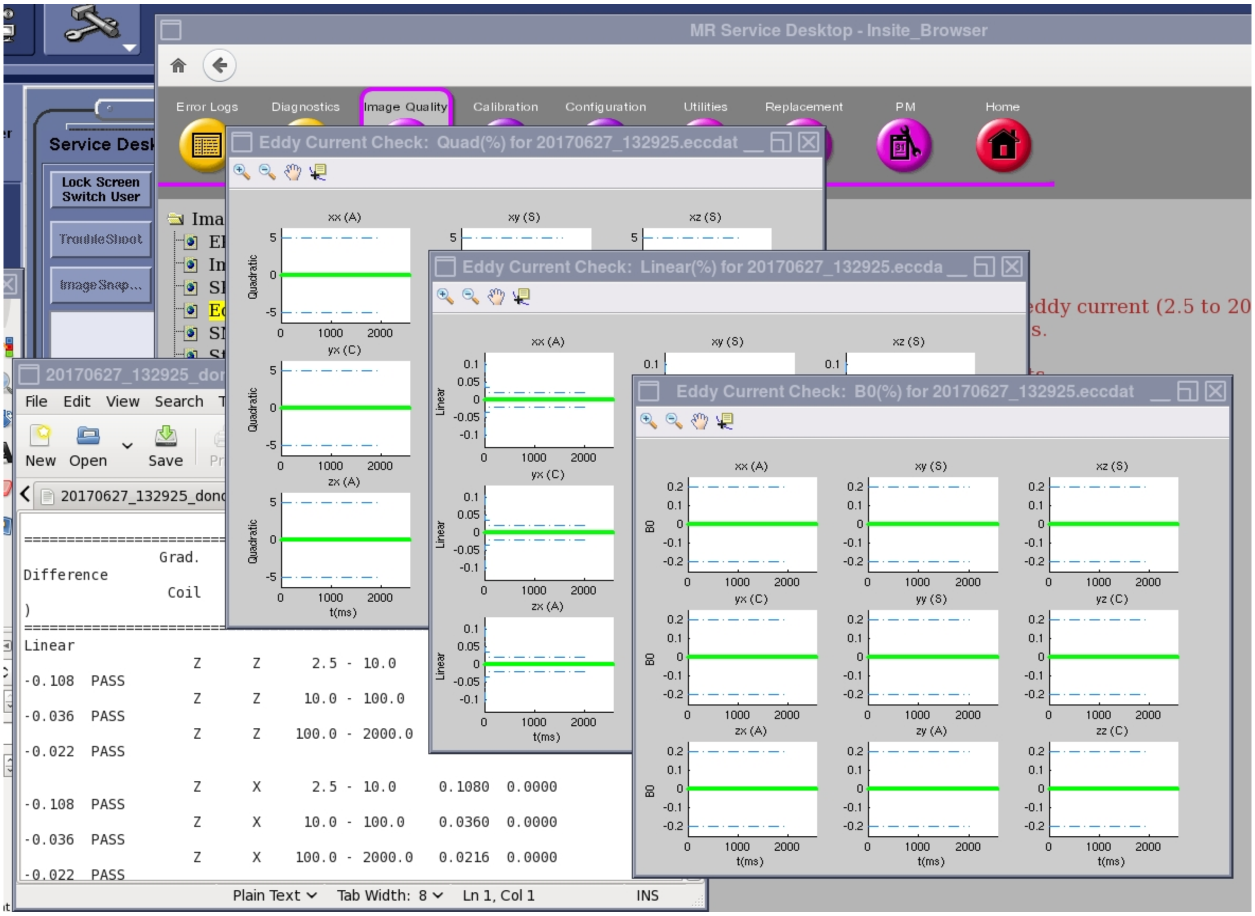
3.3.2.2 Eddy current check viewer

When you click the i button next to this tool on the SPT interface (only for PASS or FAIL status), the viewer shows. The viewer always shows the latest results when it launches.

The viewer is also available from the CSD:

- From the CSD, go to ***Image Quality*** > ***Eddy Current Check Tool***.
- Click ***Eddy Current Check Results Viewer*** to open the viewer.
- Click to open the viewer.

Figure 47 Eddy current check tool viewer



3.3.2.3 SNR tool viewer

When you click the i button next to this tool on the SPT interface (only for PASS or FAIL status), the viewer shows. The viewer always shows the latest results when it launches.

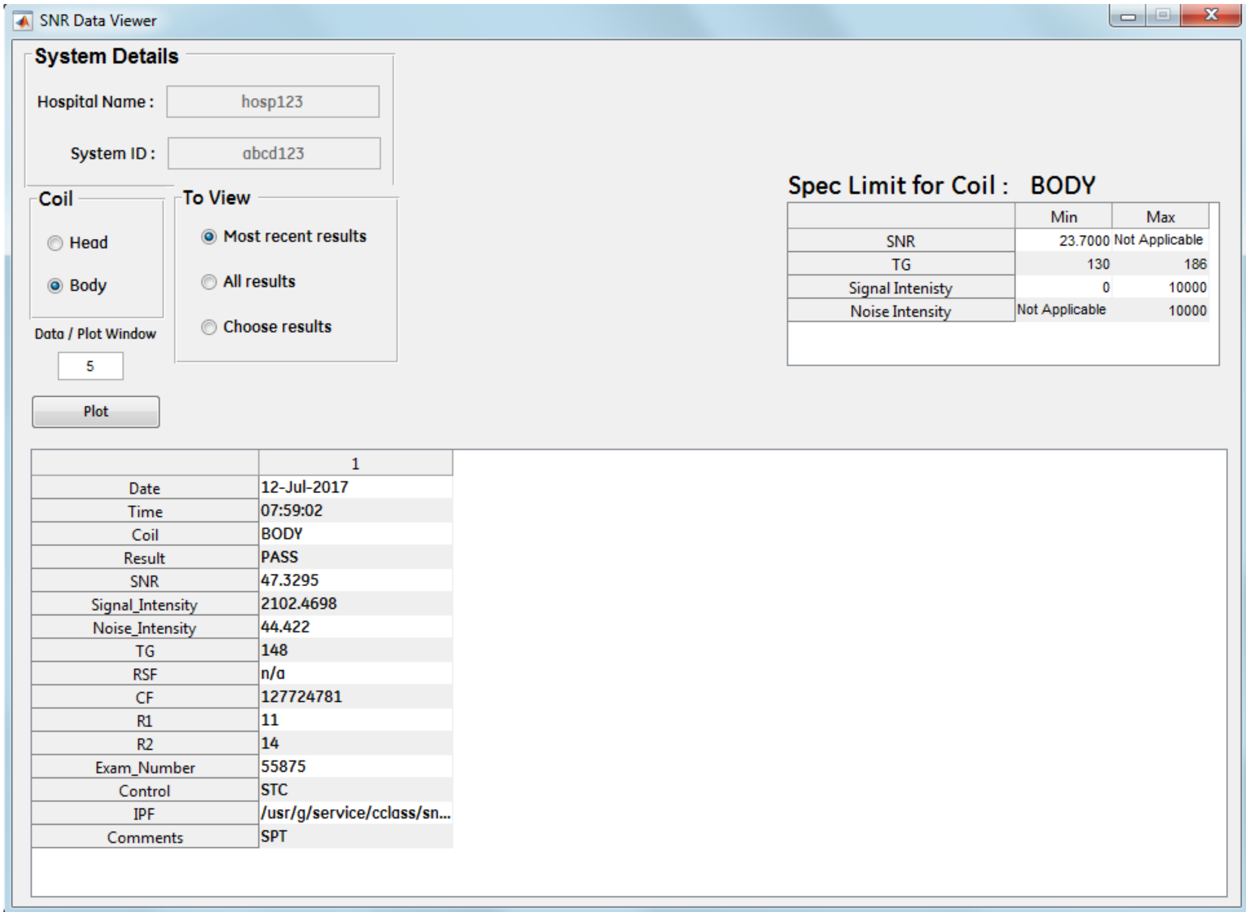
- When you click the ***Viewer*** button on the SNR Tool GUI, the viewer shows.

NOTE

Report manager is not supported for this tool. Use the Viewers to see the results in detail.

- The viewer is also available from the CSD: Go to ***Image Quality*** > ***SNR Tool***. Click ***SNR Tool Results Viewer*** to open the viewer.
- To see the most recent result, click ***Most recent result***.
- To see the overall SNR trend, click ***All results*** and then click ***Plot***.
- The spec limits show in the top right corner of the viewer.

Figure 48 SNR tool viewer

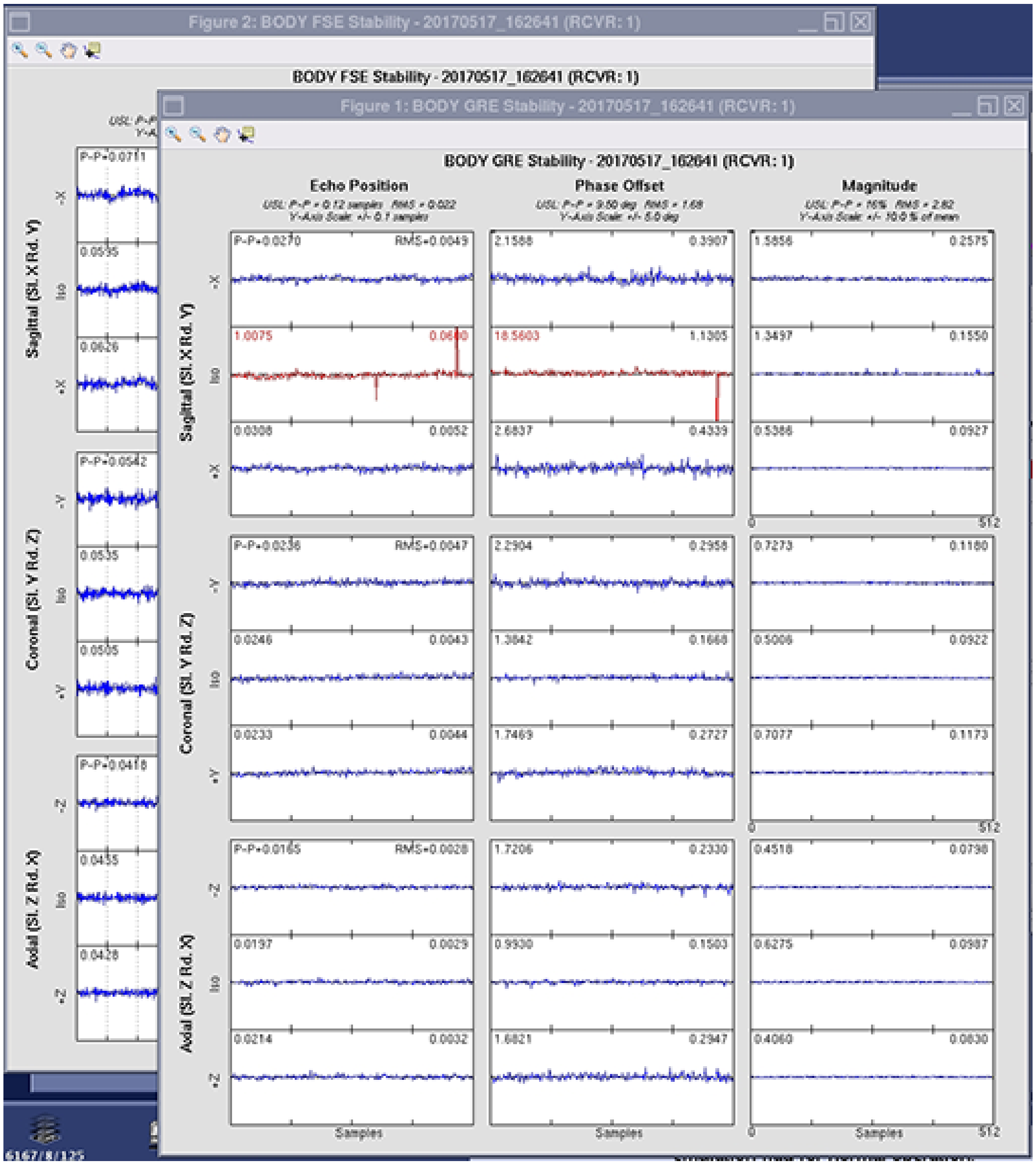


3.3.2.4 Stability tool viewer

When you click the i button next to this tool on the SPT interface (only for PASS or FAIL status), the viewer shows. The viewer always shows the latest results when it launches.

- The test results and plots will show automatically after the test is completed.
- After you close the plots, they are no longer available.

Figure 49 Stability tool output plot example



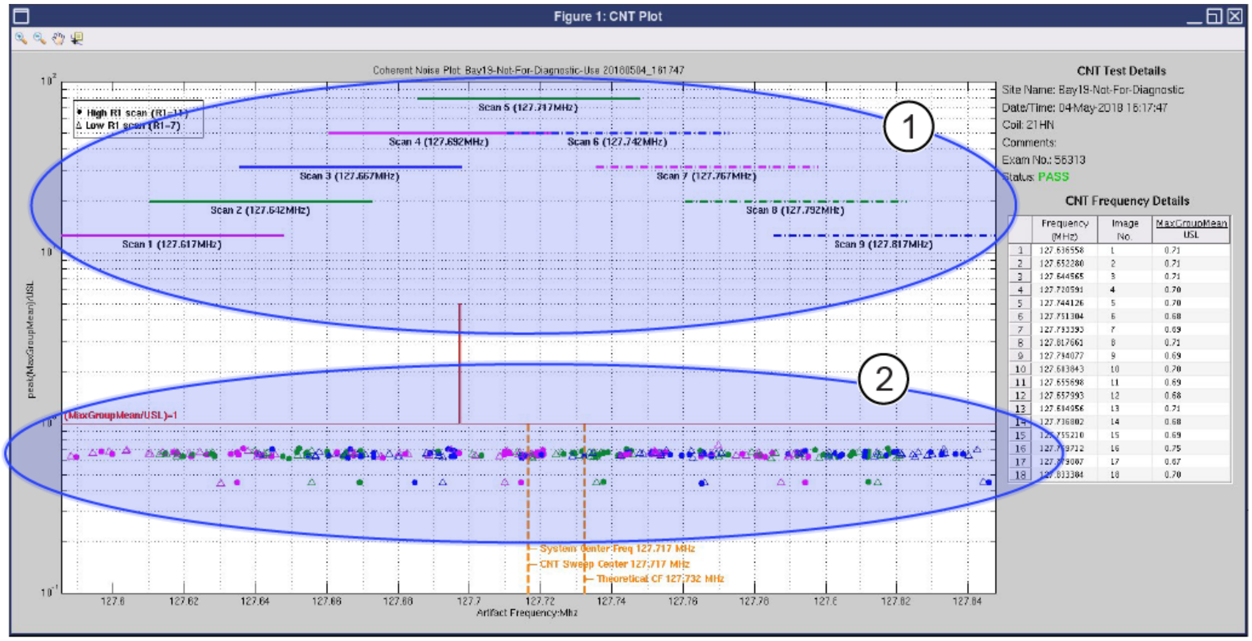
3.3.2.5 Coherent noise test tool viewer

When you click the i button next to this tool on the SPT interface (only for PASS or FAIL status), the viewer shows. The viewer always shows the latest results when it launches.

The viewer is also available from the CSD:

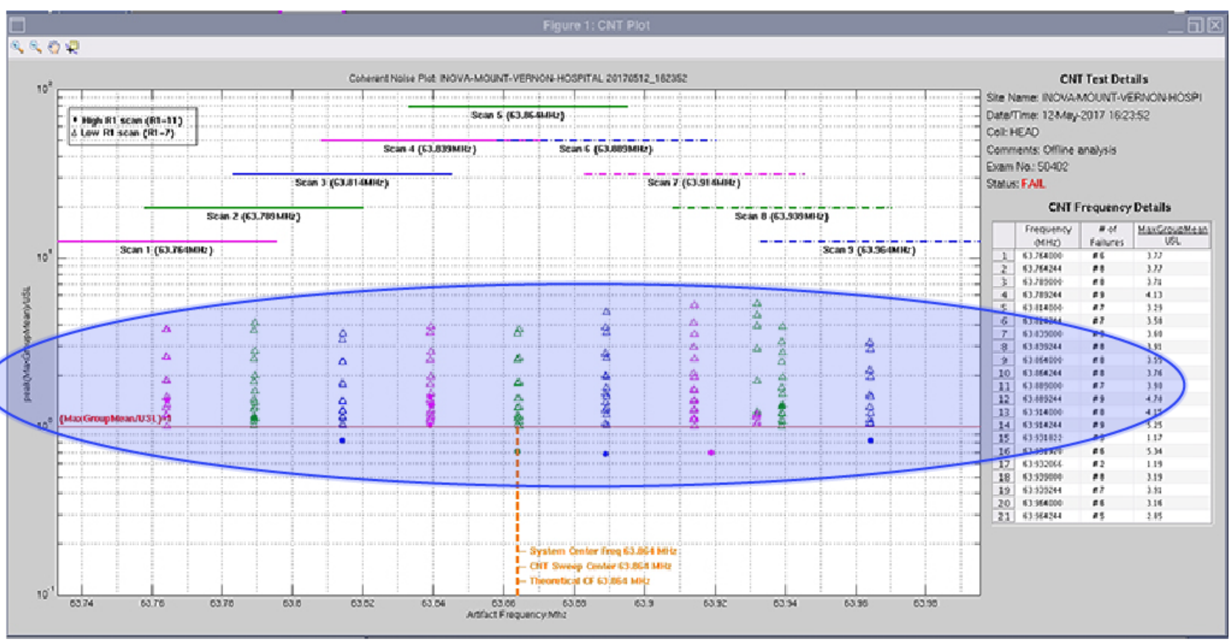
From the CSD, go to ***Image Quality*** > ***Coherent Noise Tool***. Click ***Coherent Noise Test Tool Results Viewer*** to open the viewer.

Figure 50 Coherent noise test viewer (passing results shown)



Item	Description
1	Frequency bands scanned
2	Noise within spec

Figure 51 Coherent noise test viewer: Noise out of spec (failing results shown)



3.3.2.1 System Peformance Test (SPT) user interface help

The system performance test (SPT) automatically executes Eddy Current Check, SNR, Stability and Coherent Noise tests. SPT full test mode is part of ICW and planned maintenance schedules.

3.3.2.1.1 Overview

The System Performance Test (SPT) runs phantom position checks first, and then it automatically executes the standalone versions of SPT subtools one after the other as follows:

1. Eddy Current Check: This tool measures the long term eddy current (2.5 to 2000 ms) (both B0 and linear) error for all combination of coil axes and gradient axes.
2. SNR/System Gain: This tool measures the signal-to-noise ration of a phantom. System gain is also calibrated, if it has not been completed already.
3. Stability: This tool measures and shows the magnitude drift, phase offset and echo shift using Fast Spin Echo and Gradient Resolved Echo sequences. SPT does only the body test.
4. Coherent Noise Test: This tool measures the noise that could arise due to a breach in RF shield room integrity and also zipper artifact noise.

It takes approximately two minutes for the head phantom position check and two minutes for the body phantom position check. The test will not continue unless the phantom position and setup check is successful.

All SPT tools are selected; it is not possible to deselect tools from the SPT test. You can run these tools one at a time from the Image Quality tab on the CSD.

3.3.2.1.2 SPT menu

Figure 52 SPT interface (Full Mode shown)

Table 5 SPT menu buttons

Button	Description
Start	Starts the full sequence of tests from the beginning.
Abort	Stops the SPT test temporarily.
Resume	The resume option is available for only 24 hours after you aborted. See 3.3.2.1.4 Abort and resume functions on the SPT full mode on page 66 for more details.
Exit	Closes the SPT test interface.
i	The blue information buttons next to each tool open the status for the subtools (only for PASS or FAIL status).

3.3.2.1.3 Test failures and errors

If you run SPT as part of ICW, you must correct errors or failures and then run full SPT again. All tests must pass in one uninterrupted test before you can continue ICW.

If a tool does not pass because of an error, you must click the ***Resume*** button on the Full SPT menu to continue to the next test. If a subtool fails, it will automatically start the next tool in the sequence.

3.3.2.1.4 Abort and resume functions on the SPT full mode

After SPT starts, you can stop it at any time. If you abort the test, all valid test results are assembled into a data file.

- If you click the ***Abort*** button from the ***SPT Full*** tool, this will stop the subtool that is currently in progress. The ***Exit*** button becomes active on the subtool interface. If you select ***Exit***, the ***Resume*** button becomes active on the SPT Full interface. If you select the ***Resume*** button, this will restart the subtool that was in progress when you aborted, and the full SPT test continues.
- If you click the ***Abort*** button from the subtool, this will stop the subtool that is currently in progress. The ***Exit*** button becomes active on the subtool interface. If you select ***Exit***, the ***Resume*** button becomes active on the SPT Full interface.

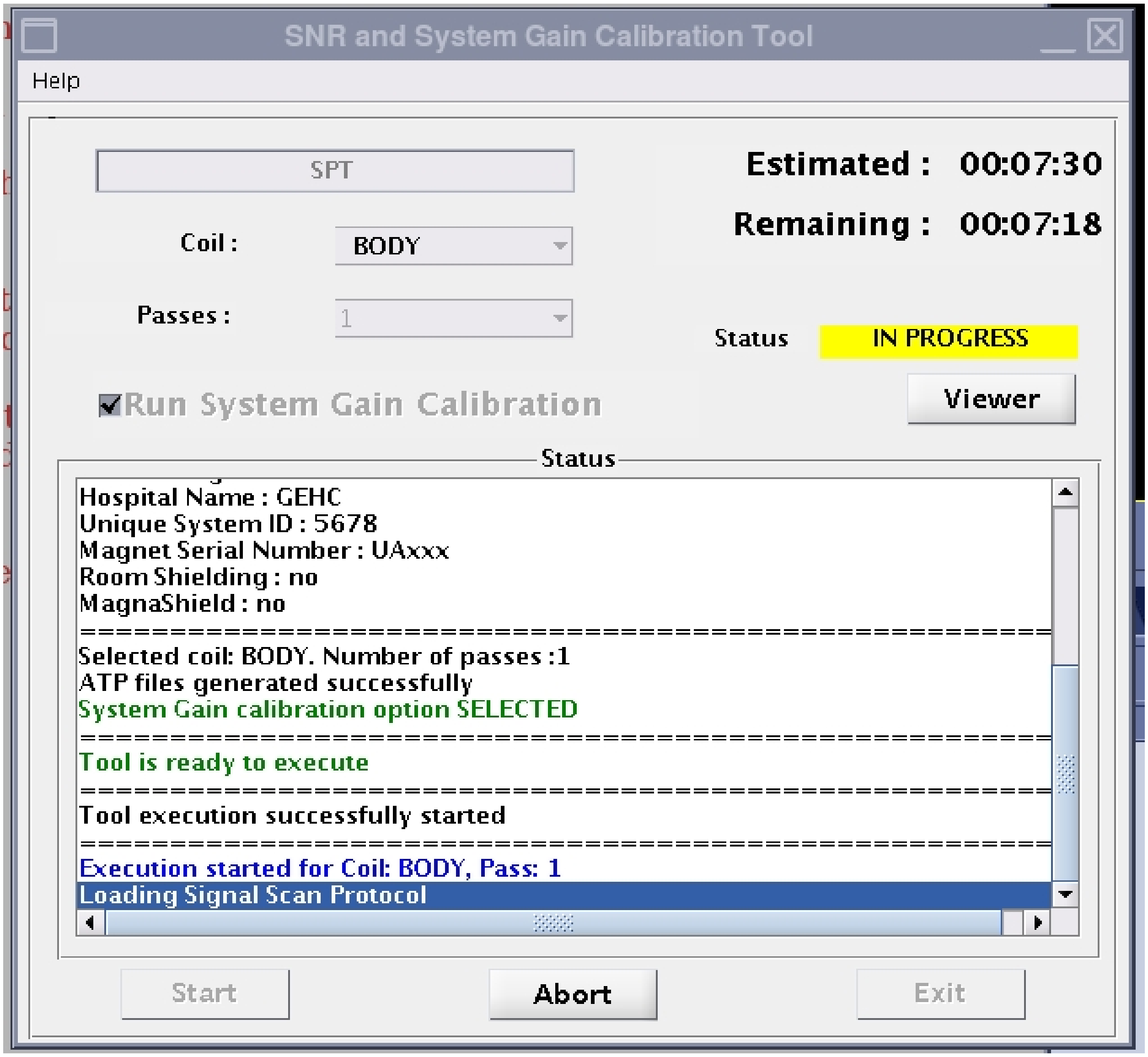
  If you click the ***Resume*** button on the SPT Full interface, this will skip the aborted subtool and will start the next subtool, and the full SPT test continues.

When you abort from the SPT Full interface, this does not skip any tests. However, if you abort SPT during ICW, this will not be considered a full run. Thus, you must click the ***Start*** button to run the full tool again and all tests must pass in order to continue ICW.

3.3.2.1.5 Subtools interface

After each subtool runs, the data results show and then close automatically. You can show them again using the blue information buttons on the SPT interface.

Figure 53 Example of a subtool interface



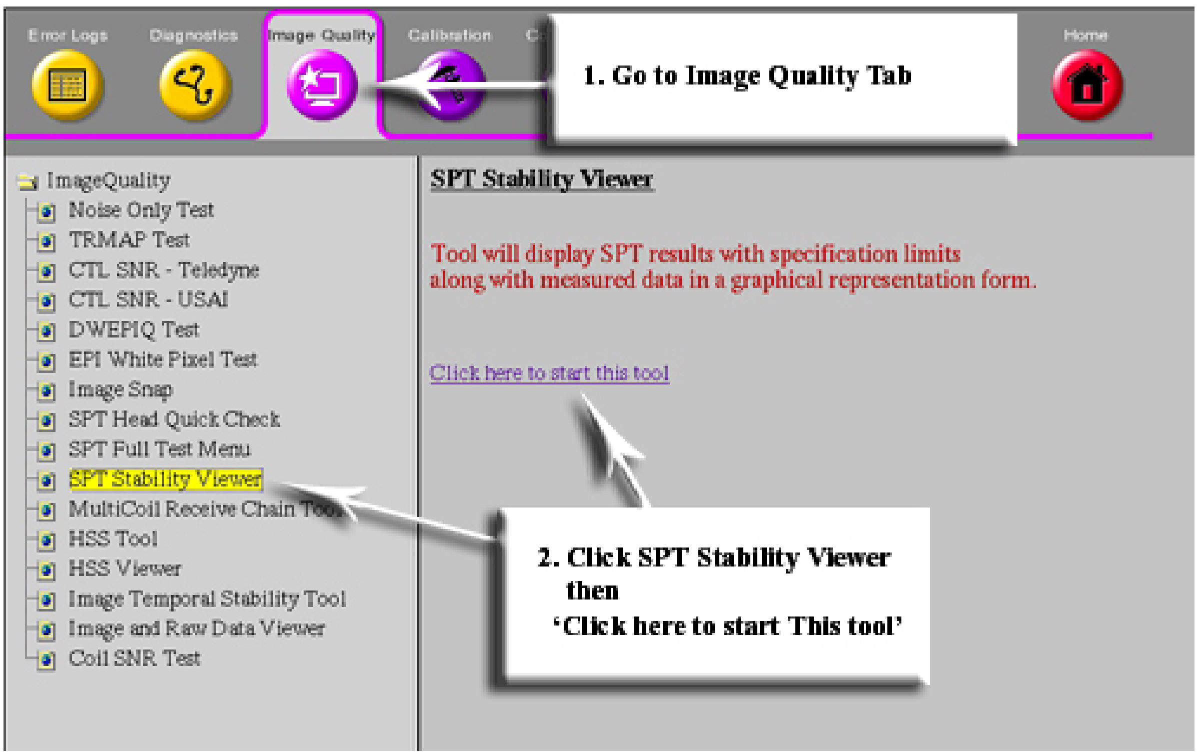
3.3.3 System Performance Test (SPT) stability viewer

Table 6 Personnel requirements

Required persons	Preliminary requirements	Procedure	Finalization
1	Not Applicable	30 minutes	Not Applicable

1. From the Common Service Desktop (CSD) select ***Image Quality*** > ***SPT Stability Viewer***.

Figure 54 Starting SPT stability viewer

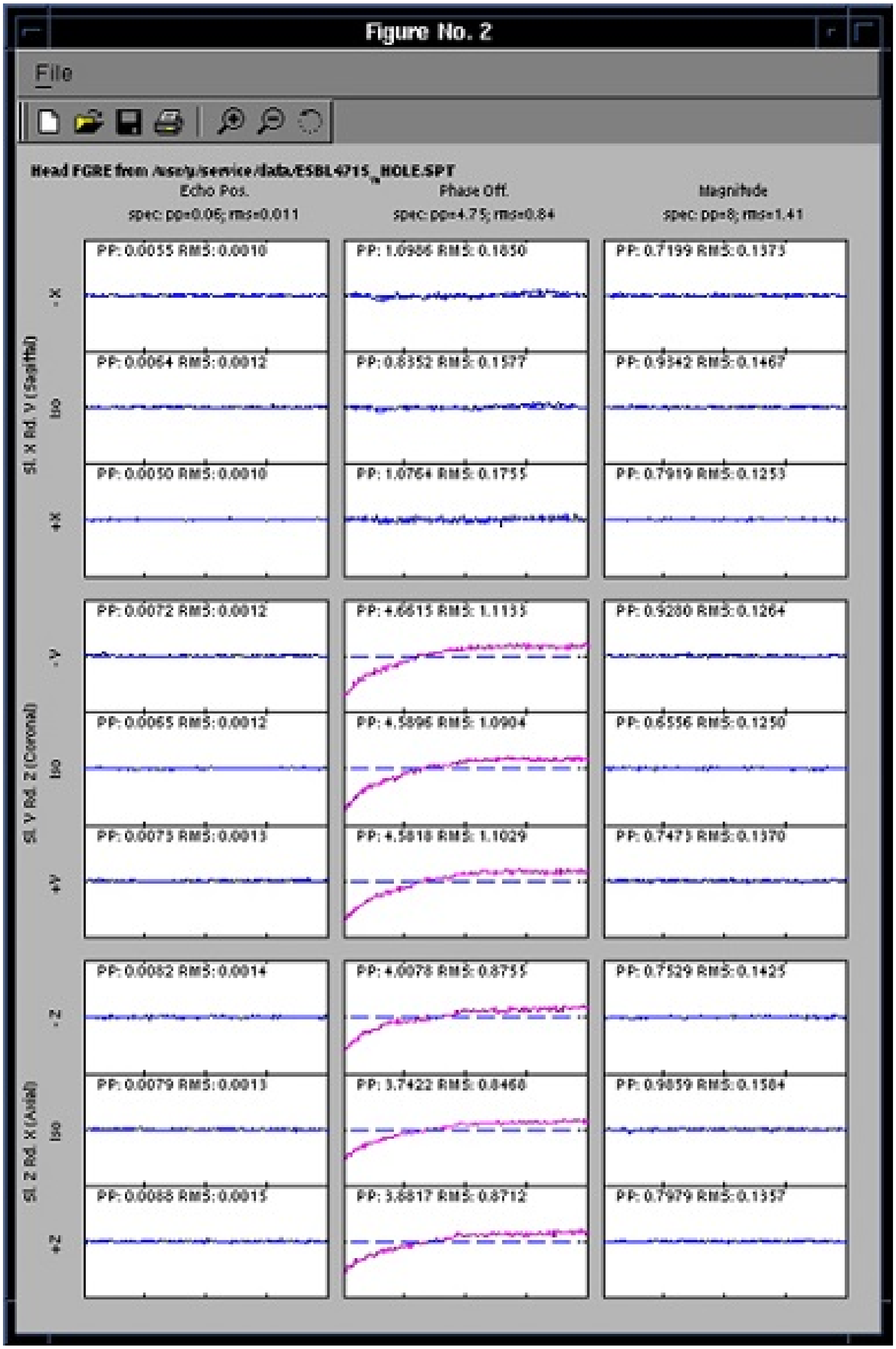


2. Select the SPT pass that you want to analyze and then click ***Open***. The most recent SPT file is at the bottom of the list.

Figure 55 Selecting file to view

3. If the selected SPT file has Stability results, then the viewer will display it. The plots for Head FSE, Head GRE, Body FSE, and Body GRE are displayed in separate windows.

Figure 56 SPT output



3.3.4 System Performance Test (SPT) test results definition

3.3.4.1 Signal-to-Noise Ratio (SNR)

In the head coil, signal is measured in the sagittal scan of the Head Quick Check set. Signal is the mean value of all pixels within two rectangular regions of interest (ROIs) in the image. Noise is measured with a no-RF scan with 30% of full scale gradient applied to each axis during data collection. Body SNR and coil gain data are collected using the SPT body sphere in its short loader. Signal is measured as in body TLT SNR. Noise data are collected in the body coil and in the head coil. In both the head and body cases, signal and noise are normalized for coil gain.

3.3.4.2 Stability tests

Stability data for the body coil is taken on each axis at seven slice locations centered about isocenter. Slices 1, 4, and 7 are processed. In the body coil case, data from the center and outermost slices are processed. In the head coil case, only data from the outermost slices are processed since there is very little signal-producing material in the axial and coronal isocenter slices of the DQA phantom.

3.3.4.3 Gradient recalled echo

- After auto prescan is completed, CVs are modified to kill the phase encoding gradient. Minimum slice thickness is used to stress the gradient subsystem.
- Time domain echo shift deviation and constant phase deviation and magnitude deviation are plotted.
- A linear ramp is fitted to and then subtracted from the plot data for each parameter prior to computing peak-to-peak values. Peak-to-peak amplitude of each linear ramp is also reported. It is important to note that plots are shown with linear ramps included, not subtracted out.

3.3.4.4 Fast spin echo

- Multi-slice stability tests using a fast spin echo clinical pulse sequence are employed by SPT. After auto prescan, CVs are modified to kill the phase-encoding gradient. Minimum slice thickness is used to stress the gradient subsystem.
- Fast spin echo data are acquired with the CV raw data set to 1 to force raw data to be stored with no preprocessing.
- Echo shift and constant phase deviation, and magnitude deviation are plotted with data arranged in order by echo (for example, all echo 1 data, followed by echo 2 data, and so on).

Doing a TPS Reset

This document provides instructions for performing a TPS reset and examples of the results output. If the system is not working properly, you can try a TPS reset to re-initialize the ISC chassis.

Personnel Requirements			
Required persons	**Preliminary require-ments**	**Procedure**	**Finalization**
1	-	15 minutes	10 minutes

Safety

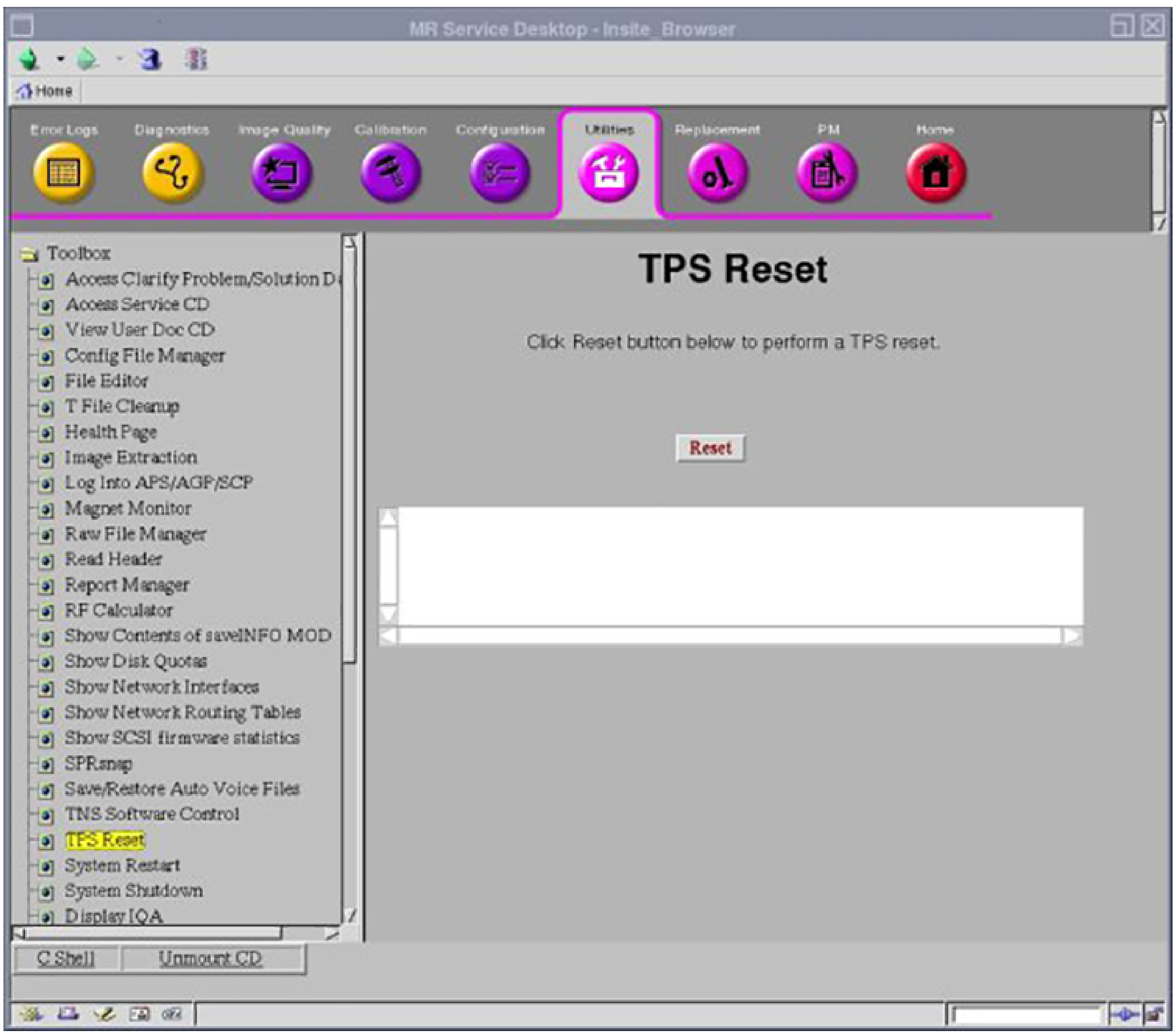
Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must:

- Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare *MR Service Safety Manual* (5452735).
- Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses).
- Comply with all site-specific training and workplace safety requirements.

If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

1. To perform a TPS reset, open the Common Services Desktop.
2. Under the ***Utilities*** tab, click ***TPS Reset*** in the left navigation pane.

Figure 57 Utilities tab



3. Click ***Reset*** to begin the reset. The following sequence of messages appears:
   - `Reset/Download TPS was successful.`
   - `Reset/Download TPS in progress.`
   - `TPS reset has completed.`

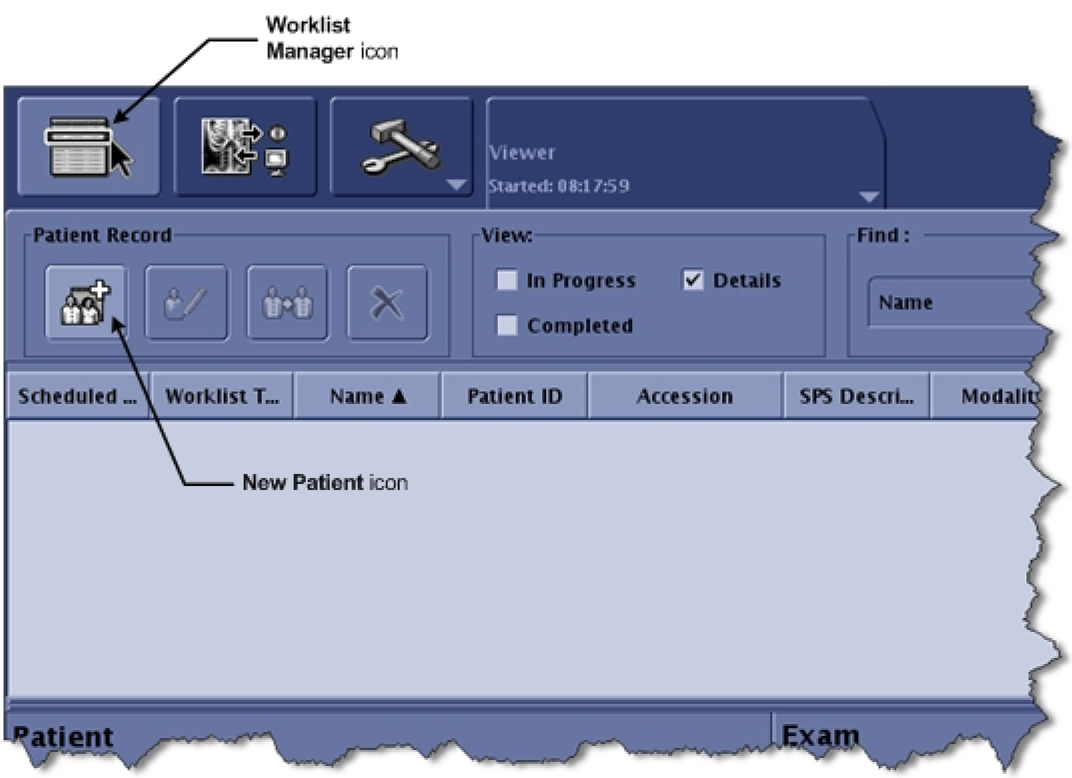
3.4 Check Scan

The following describes a workflow to follow when you need to perform a basic scan, such as a “goodbye scan” or “check scan”. It uses the 3-plane 2D localizer scan, which is relatively simple to set up and perform.

Other scans may involve additional setup and configuration. If you choose to use a different scan, consult the Operator Manual for the system you are using. The 3-plane localizer scan using the DQA phantom gives the best confidence of correct system functionality by showing correct image geometry and orientation. This is especially useful if work performed involved disconnecting and reconnecting gradient related components.

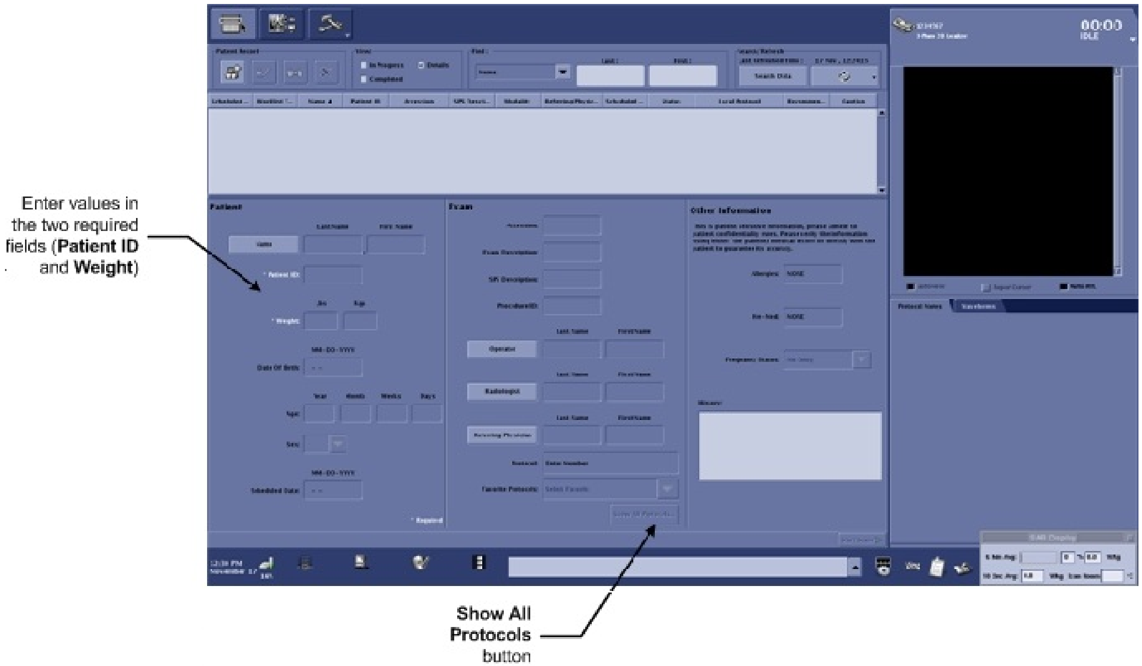
1. Place the DQA phantom in the coil.
2. Landmark on the coil/phantom.
3. At the GOC console, click the ***Worklist Manager*** icon.

Figure 58 Worklist Manager and New Patient Icons



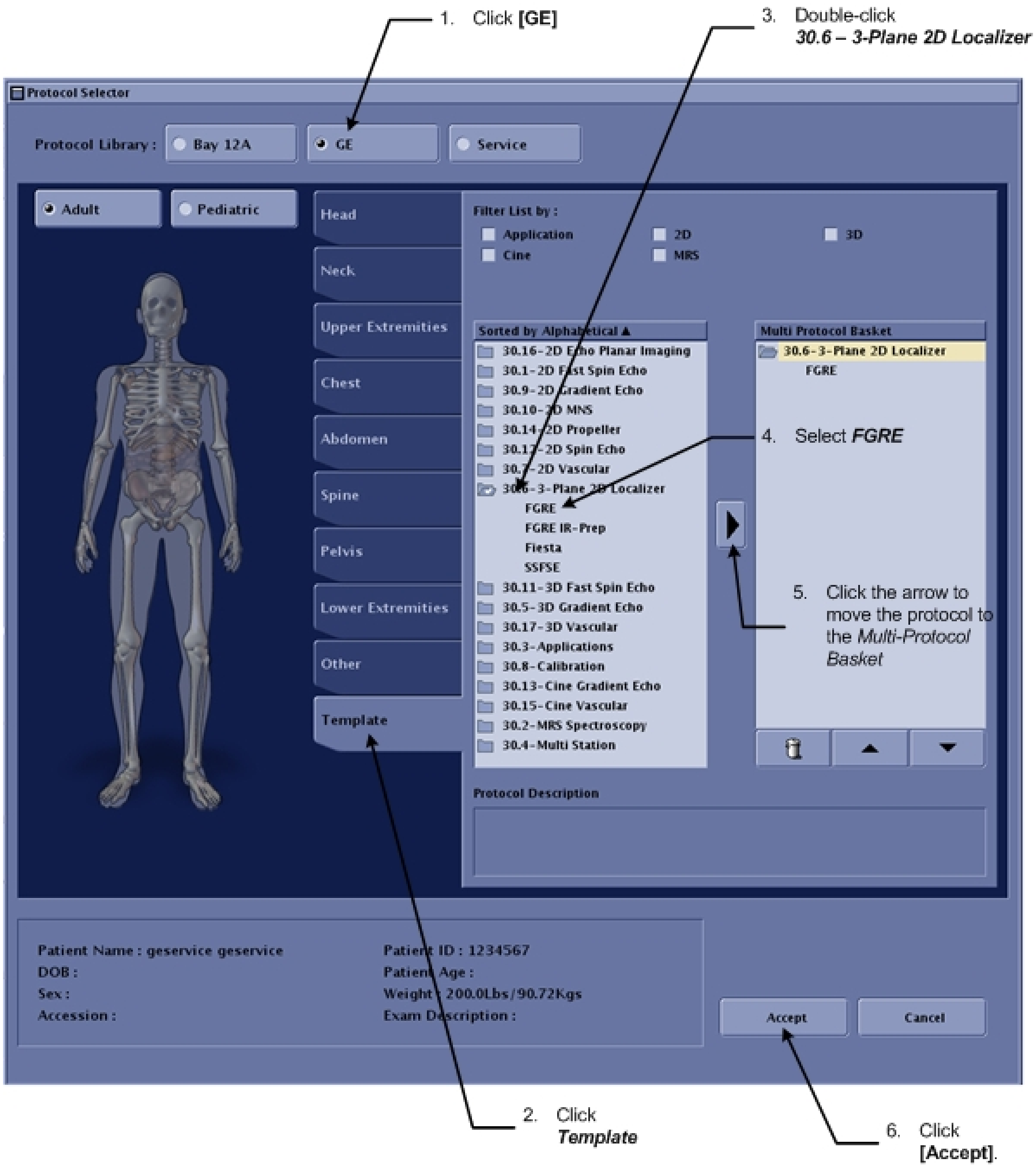
4. Click the ***New Patient*** icon.
5. In the ***Patient*** area of the screen, fill in the required fields:
    a. In the ***Patient ID*** field, enter the reason for the scan (for example, `goodbye scan`).
    b. In the ***Weight*** field, enter `200` lbs.

Figure 59 Patient Information



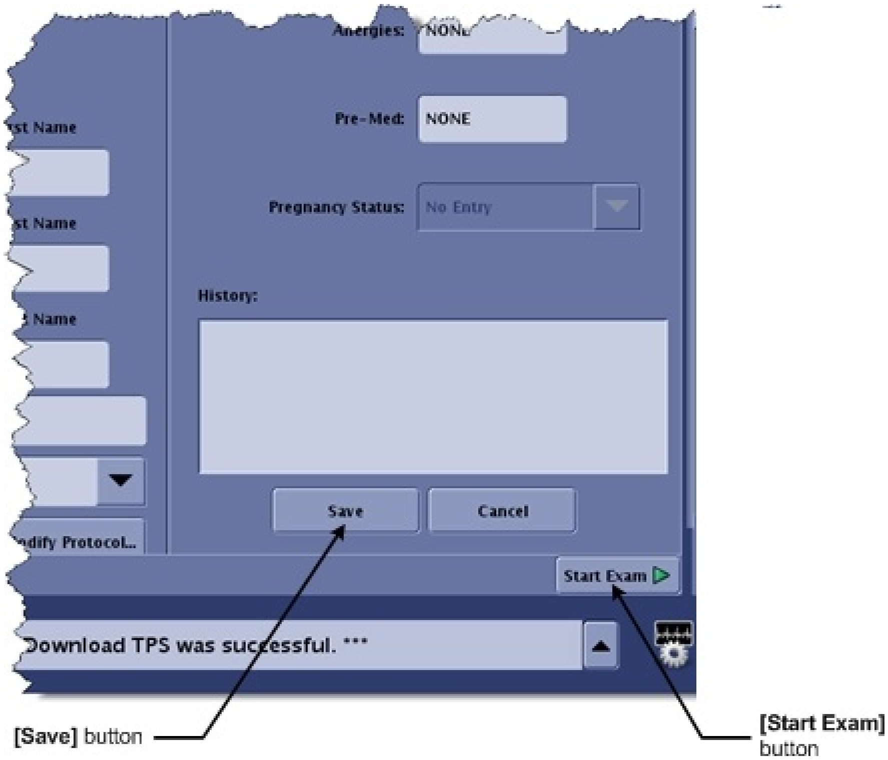
6. Click ***Show All Protocols …***.
7. On the ***Protocol Selector*** window:
    a. Click the ***GE*** button in Protocol Library.
    b. Click the ***Template*** tab.
    c. In the ***Sorted by Alphabetical*** list, double-click the ***3-Plane 2D Localizer*** folder.
    d. Select ***FGRE***.
    e. Click the right arrow to move the FGRE protocol to the ***Multi-Protocol Basket***.
    f. Click ***Accept***.

Figure 60 Protocol Selector Window



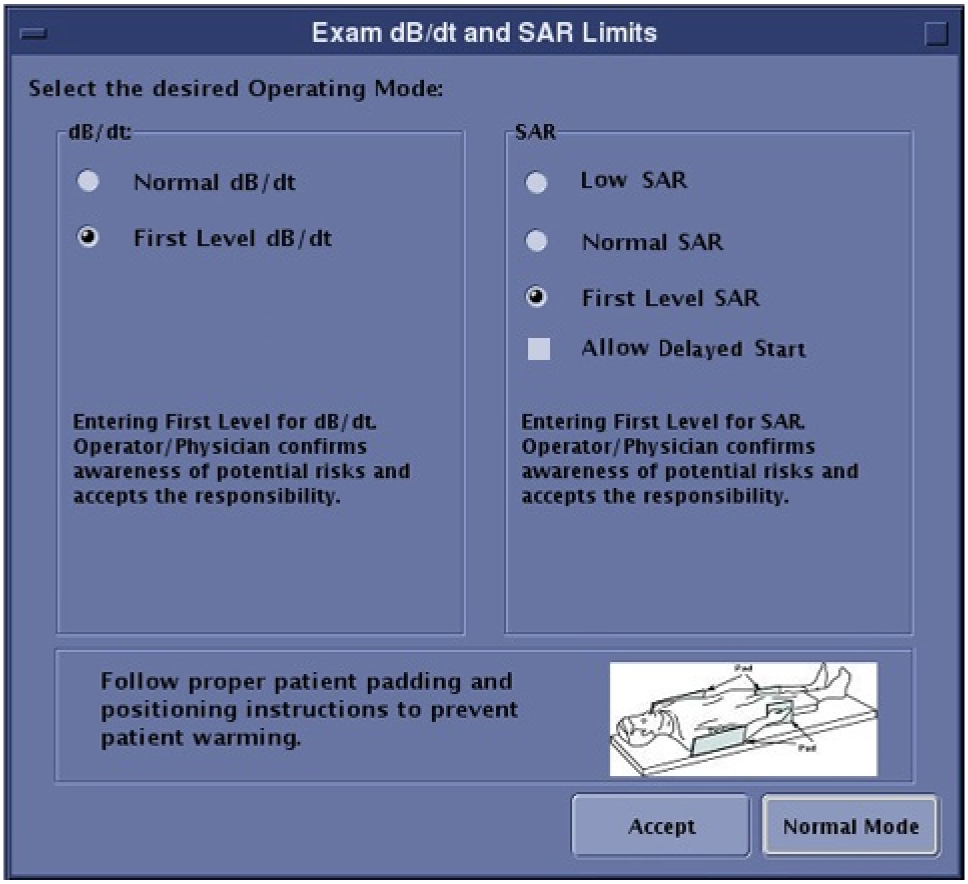
8. Click ***Save***.
9. Click ***Start Exam***.

Figure 61 Save and Start Exam Buttons



10. You may be prompted to select the dB/dt and SAR limits for the scan. If so, click ***Normal Mode***, and then click ***Accept***.

Figure 62 Exam dB/dt and SAR Levels Window



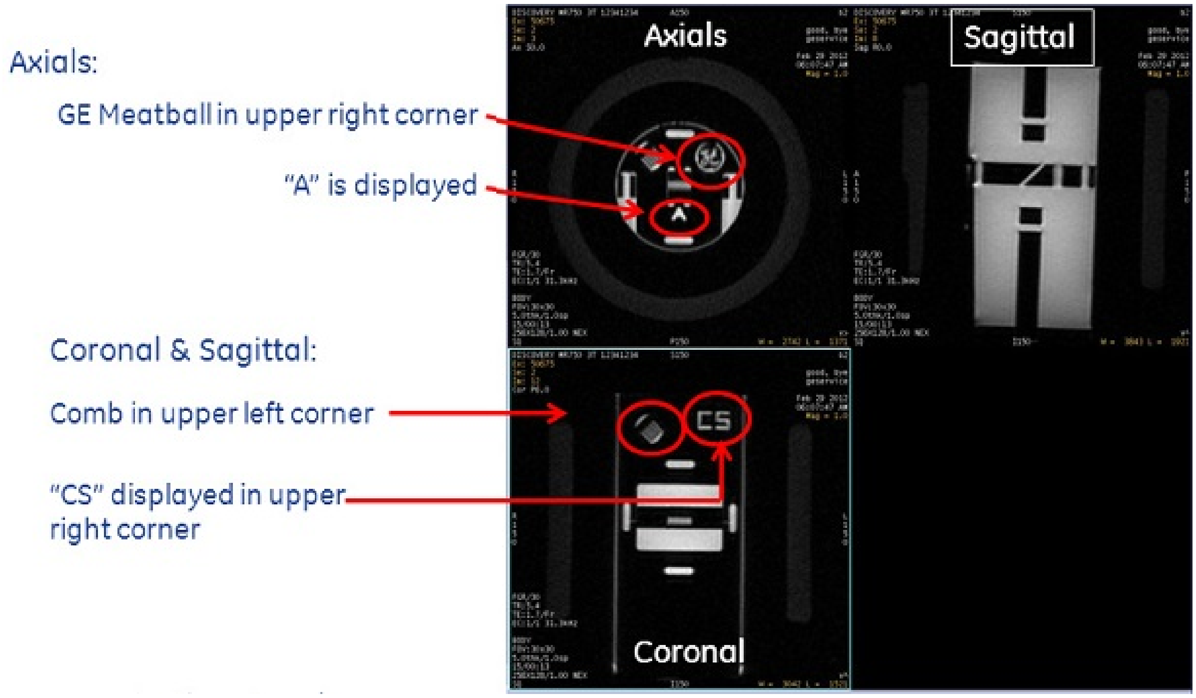
11. Click ***Coil***.
12. In the Coil Selection window, select the coil you want to use for the scan, and then click ***Accept***..

    NOTE

    The easiest way to select a coil that is connected to the system is to click the ***Currently Connected Coils*** icon, and then choose the coil from the list.

13. Because you changed the coil for the scan, the system asks you to confirm the change. Click ***Apply All***.
14. Set a value for the ***Freq FOV*** field.
15. Set a value for the ***Slice Thickness*** field.
16. Review the scan parameters for the protocol and geometric orientation.

Figure 63 Scan Parameters



17. Click the right arrow to display the scan parameter details. Review the details.
18. Press the ***Advance to Scan*** button (on the magnet) or the ***Move to Scan*** button (on the GOC console).
19. Click ***anatomical region*** button select one target to scan.
20. Click ***Save Rx***.
21. Click ***Scan***.

    The scan takes a few seconds to complete. You should see a scan image in the upper left-corner of the screen.
22. After the scan is complete, remove any phantoms and coils from the magnet bore.

SIGNA Voyager